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Reactions of enolates with carbonyl compounds: the aldol and Claisen reactions

Connections

➡ Building on

- Carbonyl compounds reacting with cyanide, borohydride, and bisulfite nucleophiles [ch6](#)
- Carbonyl compounds reacting with organometallic nucleophiles [ch9](#)
- Carbonyl compounds taking part in nucleophilic substitution reactions [ch10](#) & [ch11](#)
- How enols and enolates react with heteroatomic electrophiles such as Br_2 and NO^+ [ch20](#)
- How enolates and their equivalents react with alkylating agents [ch25](#)

Arriving at

- Reactions with carbonyl compounds as both nucleophile and electrophile
- How to make hydroxy-carbonyl compounds or enones by the aldol reaction
- How to be sure that you get the product you want from an aldol reaction
- The different methods available for doing aldol reactions with enolates of aldehydes, ketones, and esters
- How to use formaldehyde as an electrophile
- How to predict the outcome of intramolecular aldol reactions
- How esters react with enolates: the Claisen condensation
- How to acylate the enolates of esters and ketones
- How to get C-acylation and avoid O-acylation
- How to make cyclic ketones by intramolecular acylation
- Enamines in acylation reactions
- Modelling acylation on nature

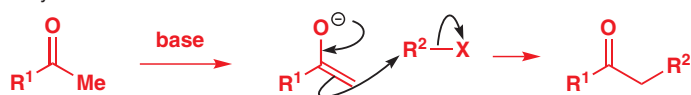
➡ Looking forward to

- Retrosynthesis [ch28](#)
- Synthesis of aromatic heterocycles [ch29](#) & [ch30](#)
- Asymmetric synthesis [ch41](#)
- Biological organic chemistry [ch42](#)

Introduction

The last chapter was about reactions of enols and enolates with alkylating agents such as alkyl halides and α,β -unsaturated carbonyl compounds. We emphasized how important it was to avoid nucleophilic attack at the carbonyl group.

alkylation of an enolate

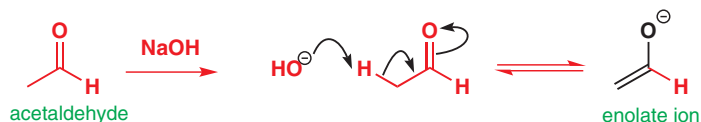


This chapter is about deliberately getting nucleophilic attack by enols and enolates on carbonyl groups of aldehydes or ketones (the aldol reaction in the first half of the chapter) or on acylating agents (the second half of the chapter).

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

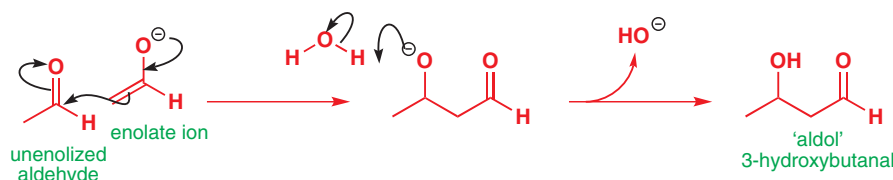
The aldol reaction

The simplest enolizable aldehyde is acetaldehyde (ethanal, CH_3CHO). What happens if we add a small amount of base, say NaOH , to this aldehyde? Some of it will form the enolate ion.



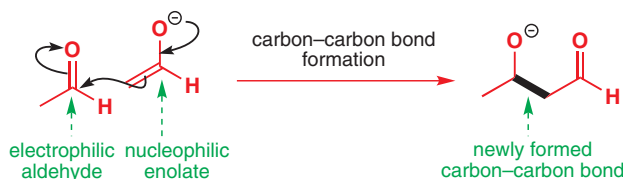
Only a small amount of the nucleophilic enolate ion is formed: as we pointed out in Chapter 25, hydroxide is not basic enough to enolize an aldehyde completely. Each molecule of enolate is surrounded by molecules of the aldehyde that are not enolized and so still have the electrophilic carbonyl group intact. The enolate ion will attack one of these aldehydes to form an alkoxide ion, which will be protonated by the water molecule formed in the first step.

■ $\text{p}K_{\text{a}} \text{H}_2\text{O} = 15.7$;
 $\text{p}K_{\text{a}} \text{MeCHO} \sim 20$



The product is an *aldehyde* with a hydroxy (*ol*) group and it has the trivial name **aldol**. The name *aldol* is given to the whole class of reactions between enolates (or enols) and carbonyl compounds even if in most cases the product is not a hydroxy-aldehyde at all. Notice that the base catalyst (hydroxide ion) is regenerated in the last step, so it is truly a catalyst.

This reaction is so important because of the carbon–carbon bond formed when the nucleophilic enolate attacks the electrophilic aldehyde. This bond is shown as a black bond in this version of the key step.

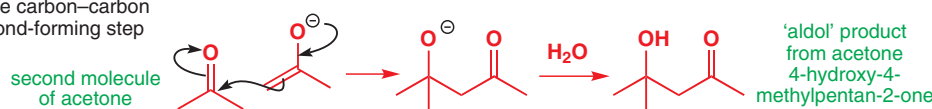


The reaction occurs with ketones as well. Acetone is a good example for us to use at the start of this chapter because it gives an important product, and as it is a symmetrical ketone, there can be no argument over which way it enolizes. Each step is the same as the aldol sequence with acetaldehyde, and the product is again a hydroxy-carbonyl compound, but this time a hydroxy-ketone.

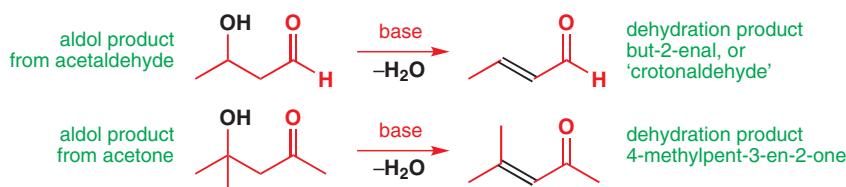
the enolization step



the carbon–carbon bond-forming step

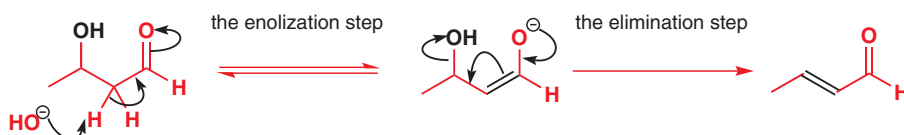


The acetaldehyde reaction works well when one drop of dilute sodium hydroxide is added to acetaldehyde. The acetone reaction is best done with insoluble barium hydroxide, $\text{Ba}(\text{OH})_2$. Both approaches keep the base concentration low. Without this precaution, the aldol products are not the compounds isolated from the reaction. With more base, further reactions occur because the aldol products dehydrate rather easily under the reaction conditions to give stable conjugated unsaturated carbonyl compounds.



■ See p. 399 for a discussion of the E1cB mechanism.

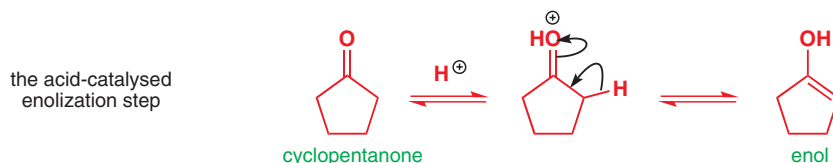
These are elimination reactions, and you met them in Chapter 17, where the possible mechanisms are discussed. You cannot normally eliminate water from an alcohol in basic solution as hydroxide is a bad leaving group. It is the carbonyl group that allows elimination here: these are E1cB reactions, with a second enolization allowing the loss of OH^- .



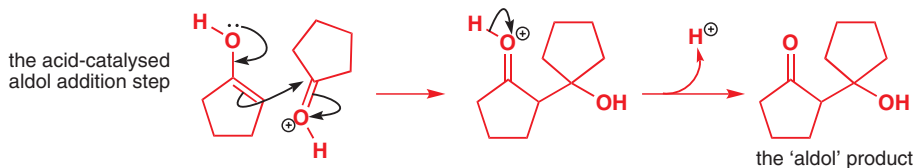
■ You do not, of course, need to learn each result individually: if you ever need to do a simple aldol reaction, you should consult the massive review in the 1968 volume of *Organic Reactions*.

In the examples that follow you will see that the base-catalysed aldol reaction sometimes gives the aldol and sometimes the elimination product. The choice is partly based on conditions—the more vigorous the conditions (stronger base, higher temperature, longer time) the more likely elimination is to occur—and partly on the structure of the reagents.

The elimination is even easier in acid solution and acid-catalysed aldol reactions commonly give unsaturated products instead of aldols. In this simple example with a symmetrical cyclic ketone, the enone is formed in good yield in acid or base. We shall use the acid-catalysed reaction to illustrate the mechanism. First the ketone is enolized under acid catalysis as you saw in Chapter 20.

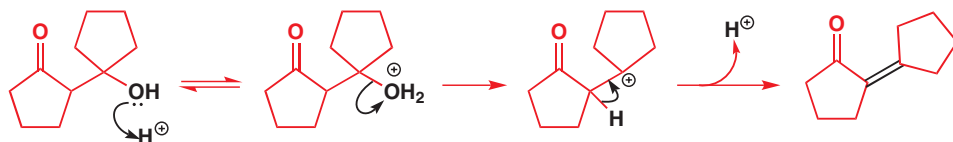


Then the aldol reaction takes place. Enols are less nucleophilic than enolates, and the reaction occurs because the electrophilic carbonyl component is protonated: the addition is acid-catalysed. An acid-catalysed aldol reaction takes place.



The aldol is a tertiary alcohol and would be likely to eliminate by an E1 mechanism in acid even without the carbonyl group. But the carbonyl ensures that only the stable conjugated enone is formed. Notice too that the dehydration is genuinely acid-catalysed as the acid reappears in the very last step.

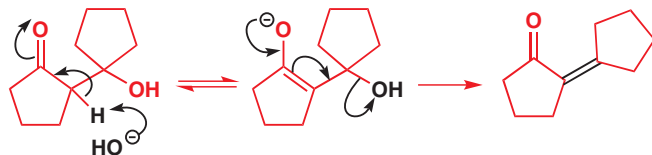
the acid-catalysed dehydration step (E1 elimination)



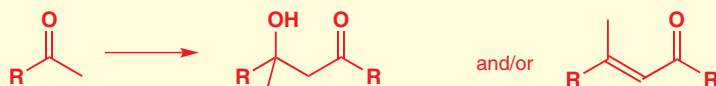
Interactive mechanism for acid-catalysed aldol reaction

None of these intermediates is detected or isolated in practice—simple treatment of the ketone with acid gives the enone in good yield. A base-catalysed reaction gives the same product via the aldol–E1cB elimination mechanism.

the base-catalysed dehydration step (E1cB elimination)



- Base-catalysed aldol reactions may give the aldol product, or may give the dehydrated enone or enal by an E1cB mechanism.
- Acid-catalysed aldol reactions may give the aldol product, but usually give the dehydrated enone or enal by an E1 mechanism.

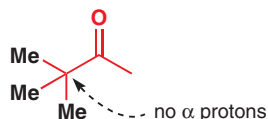


Aldol reactions of unsymmetrical ketones

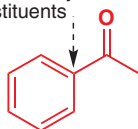
If the ketone is blocked on one side so that it cannot enolize—in other words it has no protons on that side—only one aldol reaction is possible. Ketones of this type might bear a tertiary alkyl or an aryl substituent. *tert*-Butyl methyl ketone (3,3-dimethylbutan-2-one), for example, gives aldol reactions with various bases in 60–70% yield. Enolization cannot occur towards the *t*-butyl group and must occur towards the methyl group instead.

ketones which can enolize only one way:

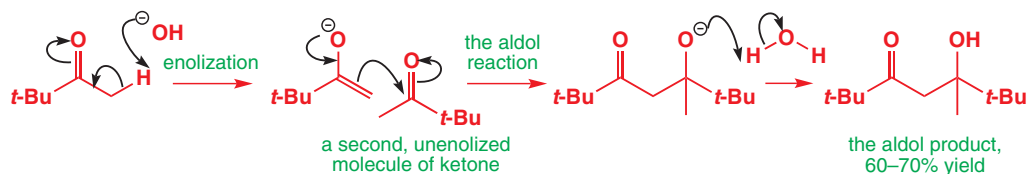
a *tert*-alkyl group cannot enolize as it has no α protons



trigonal carbon atom in aryl ring with no substituents

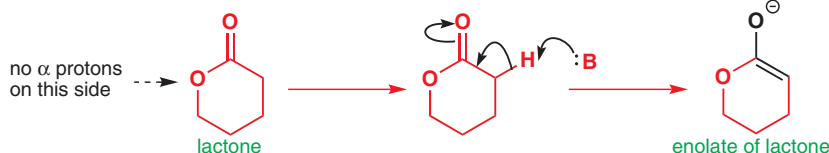


an aryl ketone has no substituents at the α position



An especially interesting case of the blocked carbonyl compound is the lactone or cyclic ester. Open-chain esters do not give aldol reactions: they prefer a different reaction that is the subject of the second half of this chapter. But lactones are in some ways quite like ketones (the stretching frequencies of their C=O groups in the IR are similar, and unlike esters they react with NaBH_4) and give unsaturated carbonyl products under basic catalysis. Enolization is unambiguous because the ester oxygen atom blocks enolization on one side.

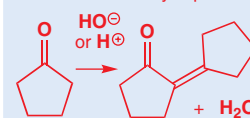
enolate formation from a lactone (cyclic ester)



Aldol condensations

The term 'condensation' is often used for reactions like this. Condensations are reactions where two molecules combine with the loss of another small molecule—usually water. In this case, two ketones combine with the loss of water. This reaction is called an aldol condensation and chemists may say 'two molecules of cyclopentanone condense together to give a conjugated enone'. You will also find the term 'condensation' used for all aldol reactions whether they occur with dehydration or not. The distinction is no longer important.

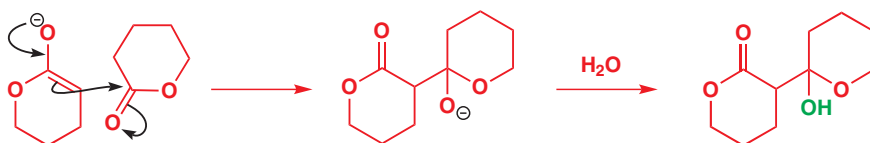
condensation of cyclopentanone



■ B in this scheme means 'base'.

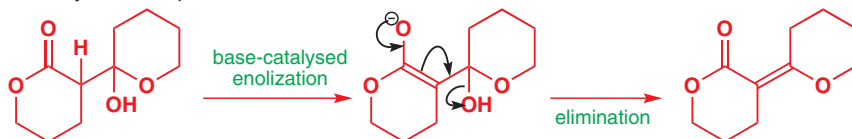
The enolate then attacks the carbonyl group of an unenolized lactone just as we have seen with aldehydes and ketones.

aldol reaction of a lactone (cyclic ester)



The last step is the familiar dehydration. As this reaction is being carried out in base we had better show the E1cB mechanism via the enolate of the aldol product.

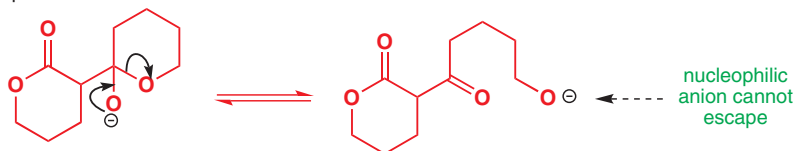
the dehydration step



You might have been surprised that the intermediate in the aldol step of this reaction did not decompose. This intermediate could be described as a tetrahedral intermediate in a nucleophilic substitution at a carbonyl group (Chapter 10). Why then does it not break down in the usual way?

The equilibrium does not affect the eventual product; it simply withdraws some of the material out of the productive reaction. We call this sort of equilibrium a *parasitic equilibrium* as it has no real life of its own—it just sucks the blood of the reaction.

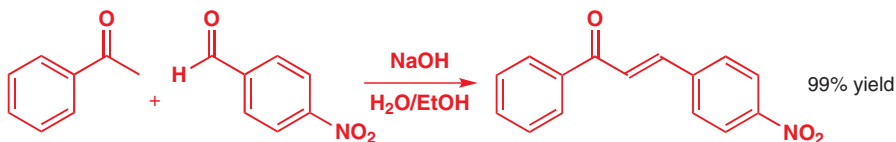
possible breakdown of a tetrahedral intermediate in a lactone aldol reaction



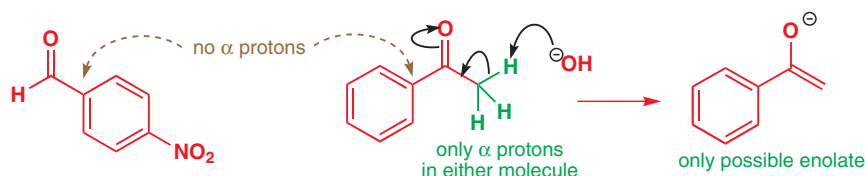
The best leaving group is the alkoxide and the product is quite reasonable. But what is it to do now? The only reasonable next step is for it to close back up again. Because the lactone is a *cyclic ester*, the leaving group cannot escape—it must stay attached to the molecule. This reaction is reversible, but dehydration is effectively irreversible because it gives a stable conjugated product. Normal, acyclic esters are different: their alkoxide leaving groups *can* leave, and the result is a different sort of reaction, which you will meet later in this chapter.

Cross-condensations

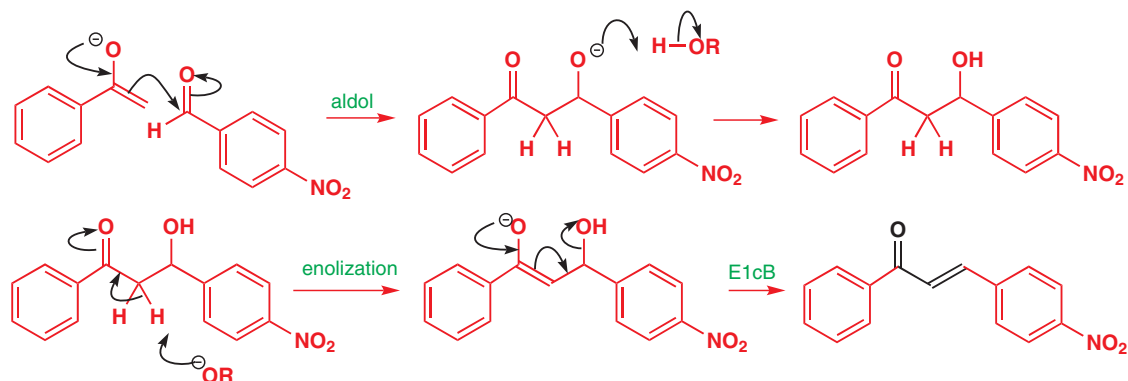
So far we have considered only 'self-condensations'—dimerization reactions of a single carbonyl compound. These form only a tiny fraction of known aldol reactions. Those that occur between two different carbonyl compounds, one acting as a nucleophile in its enol or enolate form, and the other as an electrophile, are called **cross-condensations**. They are more interesting than self-condensations, but working out what happens needs more thought. We shall start with an example that works well. The ketone PhCOMe reacts with 4-nitrobenzaldehyde in aqueous ethanol under NaOH catalysis to give a quantitative yield of an enone.



The first step must be the formation of an enolate anion using NaOH as a base. Although both carbonyl compounds are unsymmetrical, there is only one site for enolization as there is only one set of α protons, on the methyl group of the ketone. The aldehyde has no α protons at all.

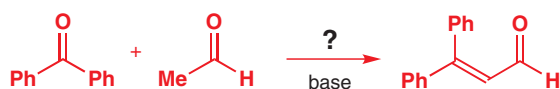


To get the observed product, the enolate obviously attacks the aldehyde to give an aldol, which then dehydrates by the E1cB mechanism.

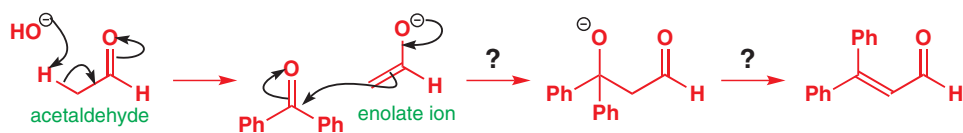


Now, in this step there was a choice. The enolate could have attacked another molecule of unenolized ketone. It didn't, because ketones are less reactive than aldehydes (Chapter 6). In this case the aldehyde has an electron-withdrawing nitro substituent too, making it even more reactive. The enolate selects the better electrophile—that is, the aldehyde.

In other cases the balance may shift towards self-condensation. You might think that a crossed aldol reaction between acetaldehyde and benzophenone (diphenylketone $\text{Ph}_2\text{C}=\text{O}$) should work well.



After all, only the aldehyde can enolize and the enolate could attack the ketone.



But it won't work. The ketone is very hindered and very conjugated. It is less electrophilic than a normal ketone and normal ketones are less electrophilic than aldehydes. Given a choice between attacking this ketone and attacking another (but unenolized) molecule of acetaldehyde, the enolate will choose the aldehyde every time. The reaction at the start of the chapter occurs, while the ketone is just a spectator.

• Successful crossed aldol reactions

For this kind of crossed aldol reaction to work well we must have two conditions:

- One partner only must be capable of enolization.
- The other partner must be incapable of enolization and be *more electrophilic than the enolizable partner*.

Everyone remembers the first of these conditions, but it is easy to forget the second.

The Mannich reaction

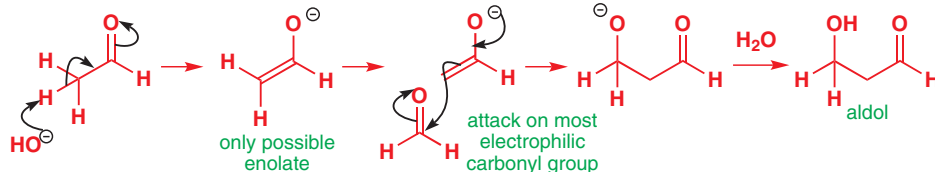
At first sight formaldehyde (methanal, $\text{CH}_2=\text{O}$) seems the ideal electrophilic partner in a mixed aldol reaction. It cannot enolize. (Usually we are concerned with α hydrogen atoms in an aldehyde. Formaldehyde does not even have α carbon atoms.) And it is a super aldehyde. Aldehydes are more electrophilic than ketones because a hydrogen atom replaces one of the alkyl groups. Formaldehyde has two hydrogen atoms.

The trouble is that it is *too* reactive. It tends to react more than once and to give extra unwanted reactions as well. You might think that condensation between acetaldehyde and formaldehyde in base would be quite simple. The acetaldehyde alone can form an enolate, and this enolate will attack the more electrophilic carbonyl group, which is formaldehyde. In each reaction the only possible enolate attacks another molecule of formaldehyde. By now you have got the idea so we simply draw the next enolate and the structure of the third aldol.

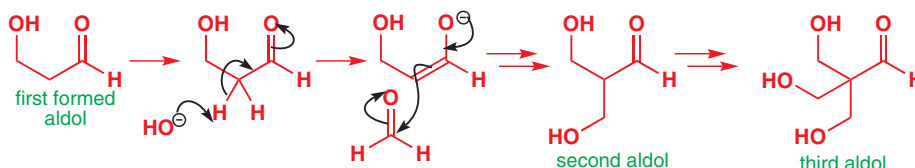
Formaldehyde

Formaldehyde is not available as a pure monomer because it forms trimers and tetramers in the pure state. The aqueous solution 'formalin' used to preserve biological specimens is available—it is 37% formaldehyde and mostly consists of the hydrate $\text{CH}_2(\text{OH})_2$; see Chapter 6. A pure dry polymer 'paraformaldehyde' is also available and was mentioned in Chapter 9. Neither of these is particularly useful in aldol reactions. The aqueous solution is used in the Mannich reaction that we describe shortly. It is possible to make the short-lived monomer and capture it with a lithium enolate, but this is not trivial experimentally.

crossed aldol reaction between acetaldehyde and formaldehyde

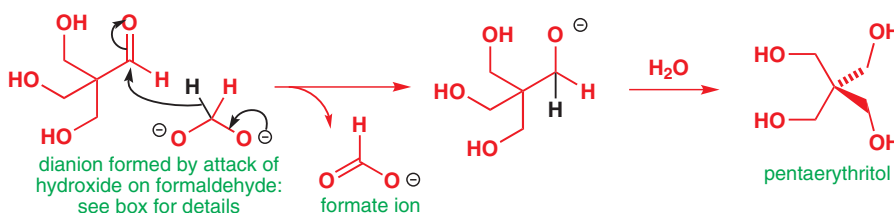


This aldol is formed all right but it is not the final product of the reaction because, with an electrophile as powerful as formaldehyde, a second and a third aldol follow swiftly on the heels of the first.



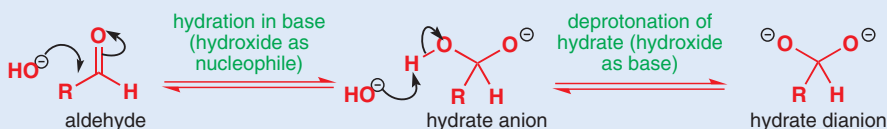
Even this is not all. A fourth molecule of formaldehyde reacts with hydroxide ion and then reduces the third aldol. This reduction is known as the Cannizzaro reaction, and is described in the box below. The final product is the highly symmetrical 'pentaerythritol', $\text{C}(\text{CH}_2\text{OH})_4$, with four CH_2OH groups joined in a tetrahedral array about the same carbon atom. The overall reaction uses four molecules of formaldehyde and can give a high yield (typically 80% with NaOH but as much as 90% with MgO) of the product.

reduction by the Cannizzaro reaction

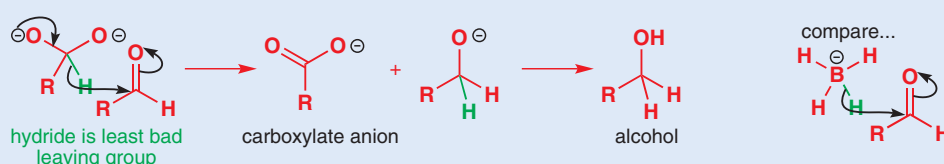


The Cannizzaro reaction

As you know, aldehydes are generally at least partly hydrated in water. Hydration is catalysed by base, and we can represent the hydration step in base like this. The hydration product is an anion but, if the base is sufficiently strong (or concentrated) and as long as the aldehyde cannot be enolized, at least some will be present as a dianion.



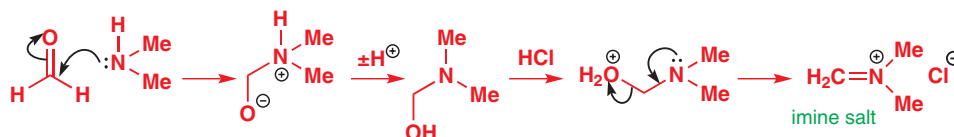
The dianion is very unstable, and one way in which it can become much more stable is by behaving like a tetrahedral intermediate. Which is the best leaving group? Out of a choice of O^{2-} , R^- , and H^- , it's H^- that (if reluctantly) has to go. Hydride is, of course, too unstable to be released into solution but, if there is a suitable electrophile at hand (another molecule of aldehyde, for example), it is transferred to the electrophilic centre in a mechanism that bears some resemblance to a borohydride reduction.



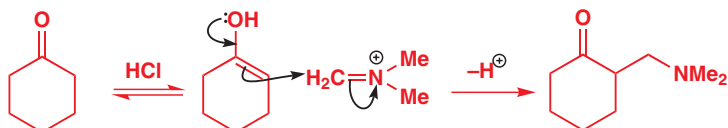
➔ There is more on the mechanism of the Cannizzaro reaction in Chapter 39.

A general solution to using formaldehyde in aldol reactions is to use the **Mannich reaction**. A typical example is shown in the margin: the reaction involves an enolizable aldehyde or ketone (here we use cyclohexanone), a secondary amine (here dimethylamine), the Mannich reaction formaldehyde as its aqueous solution, and catalytic HCl. The product is an amino-ketone from the addition of one molecule each of formaldehyde and the amine to the ketone.

The mechanism involves the preliminary formation of an imine salt from the amine and formaldehyde. The amine is nucleophilic and attacks the more electrophilic of the two carbonyl compounds available, which is, of course, formaldehyde. No acid is needed for this addition step, but acid-catalysed dehydration of the addition product gives the imine salt. In the normal Mannich reaction, this is just an intermediate but it is quite stable and the corresponding iodide is sold as Eschenmoser's salt for use in Mannich reactions.

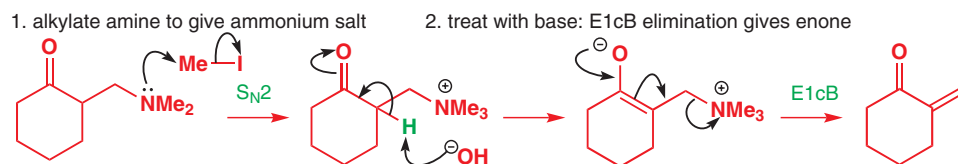


The electrophilic salt can now add to the enol (we are in acid solution) of the ketone to give the product of the reaction, an amine sometimes called a **Mannich base**.



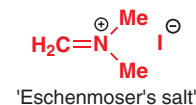
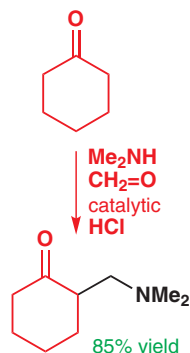
By using this reaction, you can add one molecule of formaldehyde—and one only—to carbonyl compounds. You might, of course, reasonably object that the product is not actually an aldol product at all—indeed, if you wanted the aldol product, the Mannich reaction would be of little use to you. It nevertheless remains a very important reaction. First of all, it is a simple way to make amino-ketones and many drug molecules belong to this class.

Secondly, the Mannich products can be converted to enones. The most reliable method for making the enone is to alkylate the amine product of the Mannich reaction with MeI and then treat the ammonium salt with base. Enolate ion formation leads to an E1cB reaction rather like the dehydration of aldols, but with a better leaving group.

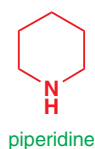
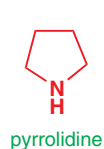


Enones like this, with two hydrogen atoms at the end of the double bond, are called **exomethylene compounds**; they are very reactive and cannot easily be made or stored. They certainly cannot be made by aldol reactions with formaldehyde alone as we have seen. The solution

the Mannich reaction



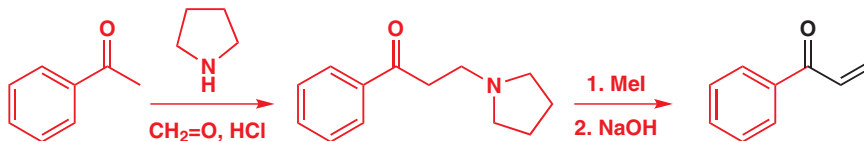
Interactive mechanism for the Mannich reaction



is to make the Mannich product, store that, and then to alkylate and eliminate only when the enone is needed. We have seen how useful this is in the Michael reaction in Chapter 25.

If the enone is wanted, any secondary amine will do as it does not end up in the molecule so the more convenient (less volatile and less smelly) cyclic amines, pyrrolidine, and piperidine, are often used. The very electrophilic enones with monosubstituted double bonds can be made in this way.

a Mannich reaction using pyrrolidine



Carbonyl compounds that are electrophilic but cannot enolize

Good crossed aldol condensations require one component to enolize and act as a nucleophile and the other not to enolize and to act as the electrophile. Here follows a list of carbonyl substituents that prevent enolization and therefore force a carbonyl compound to take the role of the electrophilic partner. They are arranged roughly in order of reactivity with the most reactive towards nucleophilic attack by an enolate at the top. You do, of course, need two substituents to block enolization so typical compounds also appear in the list. Note that the last two entries—esters and amides—do not normally do aldol reactions with enolates, but they do react as acylating agents for enolates, as you will see later in this chapter.

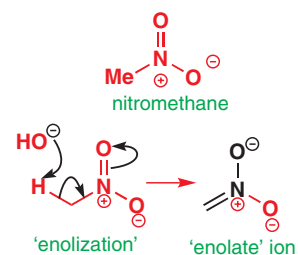
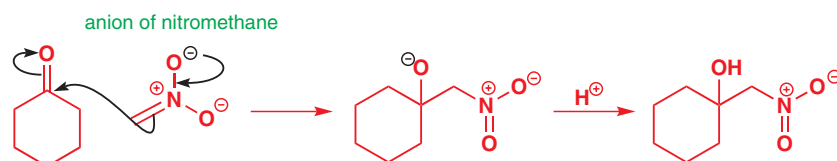
Carbonyl substituents that block enolization

	Substituent	Typical compounds	Comments
most electrophilic	H		needs special methods: see Mannich reaction
	CF ₃ , CCl ₃		made by halogenation of enols (Chapter 20)
	<i>t</i> -alkyl		many other <i>t</i> -alkyl groups
	alkenyl		nucleophile may attack alkene: see Chapter 25
	aryl		many other aromatic rings, e.g. heterocycles
	OR		formate esters and carbonates
least electrophilic	NR ₂		this is DMF: other amides unreactive

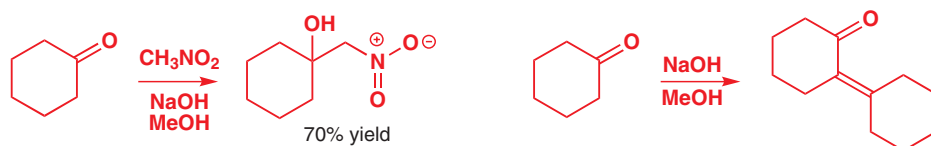
Compounds that can enolize but that are not electrophilic

We can complement this type of selectivity with the opposite type. Are there any compounds that can enolize but that cannot function as electrophiles? No carbonyl compound can fill this role, but in Chapter 25 (p. 585) we met some 'enolizable' compounds that lacked carbonyl

groups altogether. Most notable among these were the nitroalkanes. Deprotonation of nitroalkanes is not enolization nor is the product an enolate ion, but the whole thing is so similar to enolization that it makes sense to consider them together. You saw these anions, sometimes called nitronates, reacting with Michael acceptors in Chapter 25, and they also react well with aldehydes and ketones.

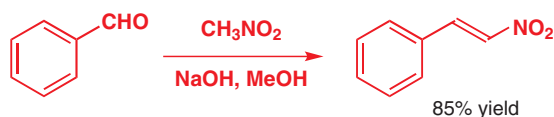


This particular example, using cyclohexanone as the electrophile and nitromethane itself as the source of the 'enolate', works quite well with NaOH as the base in methanol solution to give the 'aldol' in reasonable yield. Once again this reaction involves choice. Either compound could enolize and, indeed, cyclohexanone reacts well with itself under essentially the same conditions.



Although cyclohexanone forms an enolate in the absence of nitromethane, when both ketone and nitroalkane are present, the base prefers to remove a proton from nitromethane. This is simply a question of pK_a values. The pK_a of a typical ketone is about 20 but that of nitromethane is 10. It is not even necessary to use as strong a base as NaOH (pK_a of $H_2O = 15.7$) to deprotonate nitromethane: an amine will do (pK_a of $R_2NH_2^+$ about 10) and secondary amines are often used.

The elimination step also occurs easily with nitro compounds and is difficult to prevent in reactions with aromatic aldehydes. Now you can see how the useful nitroalkene Michael acceptors in Chapter 22 were made.



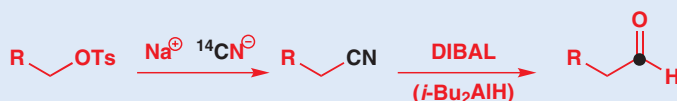
Nitroalkenes as termite defence compounds

Termites are social insects, and every species has its own 'soldier' termites that defend the nest. Soldier termites of the species *Prothinotermes simplex* have huge heads from which they spray a toxic nitroalkene on their enemies.

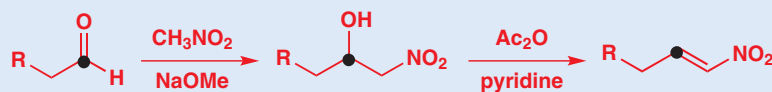
defensive nitroalkene from termite soldiers



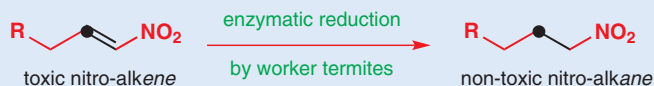
Although this compound kills other insects and even other species of termites, it has no effect on the workers of the same species. To find out why this was so, Prestwich made some radioactive compound using the aldol reaction. First, the right aldehyde was made using an S_N2 reaction with radioactive (^{14}C) cyanide ion on a tosylate followed by DIBAL reduction (Chapter 23) of the nitrile. The position of the ^{14}C atom in each compound is shown in black.



Then the aldol reaction was carried out with nitromethane and acetic anhydride in pyridine to give the nitro aldol. Elimination using sodium methoxide gave the defence compound (*E*-1-nitropentadec-1-ene) in 37% yield over the four steps.



If the worker termites were sprayed with the labelled compound, they were able to make it harmless by using an enzyme to reduce the nitroalkene to a nitroalkane. The still radioactive labelled nitroalkane could be re-isolated only from workers of the same species: other insects do not have the enzyme.



If an aldol reaction can be done with:

- only one enolizable component
- only one set of enolizable protons
- a carbonyl electrophile more reactive than the compound being enolized

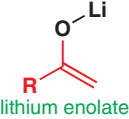
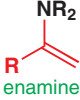
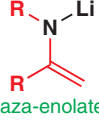
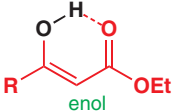
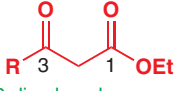
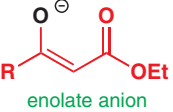
then you are lucky and the crossed aldol method will work. But most aldol reactions aren't like this: they are cross-condensations of aldehydes and ketones of various reactivities with several different enolizable protons. Crossed aldols on most pairs of carbonyl compounds lead to hopeless mixtures of products. In all cases that fail to meet these three criteria, a specific enol equivalent will be required: one component must be turned quantitatively into an enol equivalent, which will be reacted in a separate step with an electrophile. That is what the next section is about—and you will find that some of the methods have a lot in common with those we used for alkylating enolates in Chapter 25.

■ You will see exceptions to this generalization when we discuss asymmetric synthesis in Chapter 41.

Specific enol equivalents can be used to control aldol reactions

In Chapter 25 we saw that the alkylation of enolates was most simply controlled by preparing a specific enol equivalent from the carbonyl compound. The same approach is the most powerful of all the ways to control the aldol reaction. The table is a reminder of some of the most useful of these specific enol equivalents.

Important specific enol equivalents

	 enol	 carbonyl compound	 enolate ion
oxygen derivatives:	 silyl enol ether		 lithium enolate
nitrogen derivatives:	 enamine		 aza-enolate
1,3-dicarbonyls:	 enol	 1,3-dicarbonyl compound	 enolate anion

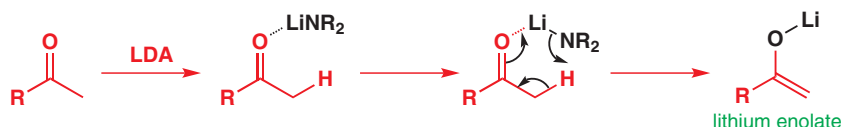
Specific enol equivalents are intermediates that still have the reactivity of enols or enolates but are stable enough to be prepared in good yield from the carbonyl compound. That was all we needed to know in Chapter 25. Now we know that a further threat is the reaction of the partly formed enol derivative with its unenolized parent and we should add that ‘no aldol reaction should occur during the preparation of the specific enol equivalent’.

● Specific enol equivalents are intermediates that still have the reactivity of enols or enolates but are stable enough to be prepared in good yield from the carbonyl compound *without any aldol reaction*.

Sensible choice of an appropriate specific enol equivalent will allow almost any aldol reaction to be performed successfully. The first two compounds in our list, the silyl enol ethers and the lithium enolates, have a particularly wide application and we should look first at the way these work. As the table suggests, silyl enol ethers are more like enols: they are non-basic and not very reactive. Lithium enolates are more like enolate anions: they are basic and reactive. Each is appropriate in different circumstances.

Lithium enolates in aldol reactions

Lithium enolates are usually made at low temperature in THF with a hindered lithium amide base (often LDA) and are stable under those conditions because of the strong O–Li bond. The formation of the enolate begins with Li–O bond formation before the removal of the proton from the position by the basic nitrogen atom.

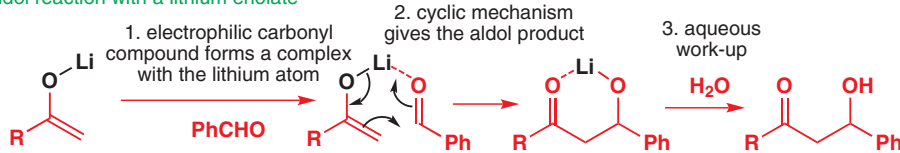


This reaction happens very quickly—so quickly that the partly formed enolate does not have a chance to react with unenolized carbonyl compound before proton removal is complete.



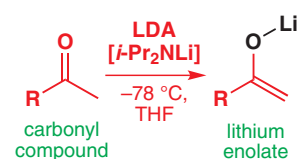
Now, if a second carbonyl compound is added, it too complexes with the same lithium atom. This allows the aldol reaction to take place by a cyclic mechanism in the coordination sphere of the lithium atom. The aldol step itself is now a very favourable intramolecular reaction with a six-membered cyclic transition state. The product is initially the lithium alkoxide of the aldol, which gives the aldol on work-up.

aldol reaction with a lithium enolate



This reaction works well even if the electrophilic partner is an enolizable aldehyde. In this example, an unsymmetrical ketone (blocked on one side by an aromatic ring) reacts as the enol partner in excellent yield with a very enolizable aldehyde. This is the first complete aldol reaction we have shown you using a specific enol equivalent: notice the important point that it is done in two steps:

- first, form the specific enol equivalent (here, the lithium enolate at low temperature)
- then add the electrophile.

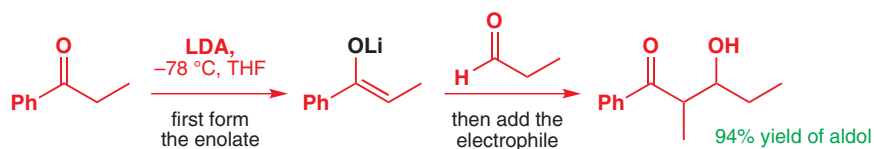


➔ The formation of lithium enolates was discussed in Chapter 25.

 Interactive mechanism for lithium enolate formation

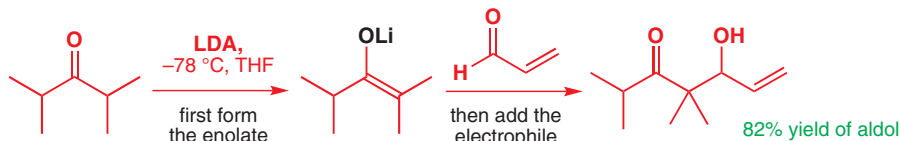
■ Aldehydes are an exception. You can make lithium enolates from some aldehydes such as *i*-PrCHO, but generally self-condensation is too fast, so unwanted aldol self-condensation products are produced during the formation of the lithium enolate. To make specific enolates of aldehydes we need to use another type of derivative: see later.

■ A lithium cation has four coordination sites—those we do not show are occupied by solvent molecules. Before the aldol reaction can take place, one of these molecules must be displaced by the electrophilic carbonyl partner.



Contrast the crossed aldols earlier in the chapter, where enolizable component, base, and electrophile were all mixed together in one step.

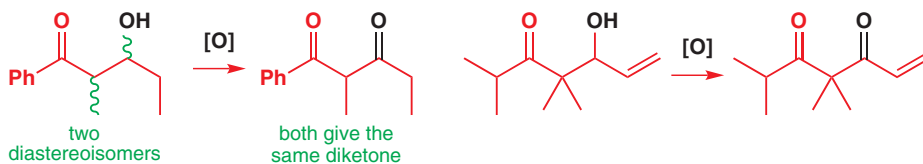
The next example is particularly impressive. The enol partner is a symmetrical ketone that is very hindered—there is only one hydrogen on either side. The electrophilic partner is a conjugated enal that is not enolizable but that might accept the nucleophile in a conjugate manner. In spite of these potential problems, the reaction goes in excellent yield.



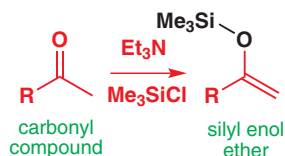
■ Because of the six-membered ring mechanism for the addition, lithium enolates don't usually do conjugate additions. For enol equivalents that do, see Chapter 25.

■ The symbol [O] denotes oxidation by one of the very general but ill-defined oxidizing agents from the laboratory of the famous Welsh chemist Owen Bracketts. Here the Swern oxidation was the best (see Chapter 23).

You may wonder why we did not mention the stereochemistry of the first of these two products. Two new stereogenic centres are formed and the product is a mixture of diastereoisomers. In fact, both of these products were wanted for oxidation to the 1,3-diketone so the stereochemistry is irrelevant. This sequence shows that the aldol reaction can be used to make diketones too.

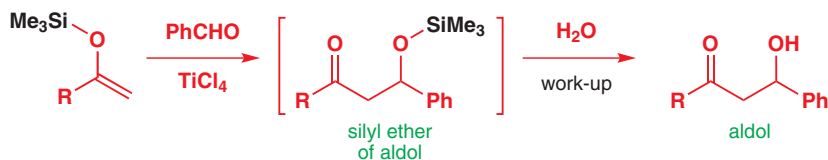


Silyl enol ethers in aldol reactions



The silyl enol ether can be prepared from its parent carbonyl compound by forming a small equilibrium concentration of enolate ion with weak base such as a tertiary amine and trapping the enolate with the very efficient oxygen electrophile Me_3SiCl . The silyl enol ether is stable enough to be isolated but is usually used immediately without storing.

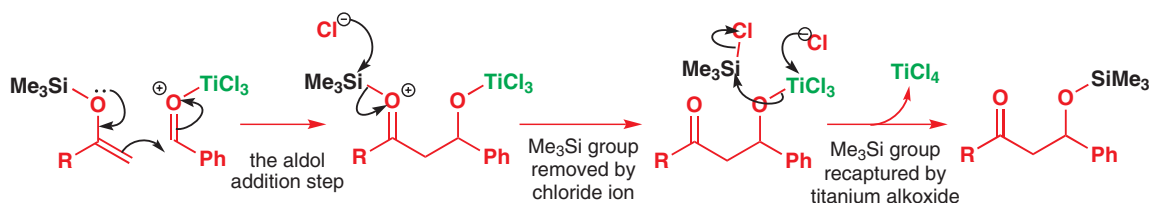
You should look upon silyl enol ethers as rather reactive alkenes that combine with things like protons or bromine (Chapter 20), but do not react with aldehydes and ketones without catalysis: they are much less reactive than lithium enolates. As with alkylation (pp. 595 and 609), a Lewis acid catalyst is needed to get the aldol reaction to work, and a Ti(IV) compound such as TiCl_4 is popular.



The immediate product is actually the silyl ether of the aldol product but this is hydrolysed during work-up and the aldol is formed in good yield. The Lewis acid presumably bonds to the carbonyl oxygen atom of the electrophile.

Now the aldol reaction can occur: the positive charge on the titanium-complexed carbonyl oxygen atom makes the aldehyde reactive enough to be attacked even by the not very nucleophilic silyl enol ether. Chloride ion removes the silyl group and the titanium alkoxide

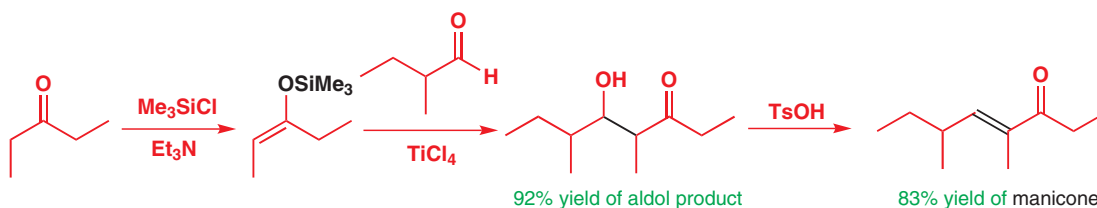
captures it again. This last step should not surprise you as any alkoxide (MeOLi, for example) will react with Me₃SiCl to form a silyl ether.



This mechanism looks complicated, and it is. It is, in fact, not clear that the details of what we have written here are right: the titanium may well coordinate to *both* oxygens throughout the reaction, and some of the steps that we have represented separately probably happen simultaneously. However, all reasonable mechanisms will agree on two important points, which you must understand:

- Lewis acid is needed to get silyl enol ethers to react.
- The key step is an aldol reaction of the silyl enol ether with the Lewis-acid complexed electrophile.

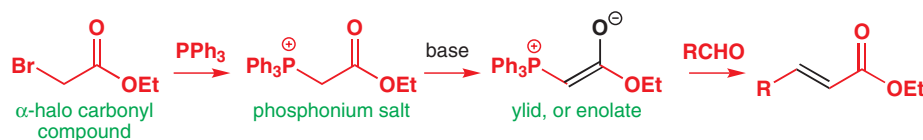
The use of silyl enol ethers can be illustrated in a synthesis of manicone, a conjugated enone that ants use to leave a trail to a food source. It can be made by an aldol reaction between pentan-3-one (as the enol component) and 2-methylbutanal (as the electrophile). Both partners are enolizable so we shall need to form a specific enol equivalent from the ketone. The silyl enol ether works well. The aldol product will be a mixture of diastereoisomers but it eliminates to give a single compound.



The silyl enol ether is not isolated but is treated immediately with the aldehyde to give an excellent yield of the aldol. Dehydration in acid solution with toluenesulfonic acid (TsOH) gives the enone. You can see by the high yield in the aldol reaction that there is no significant self-condensation of either partner in the aldol reaction.

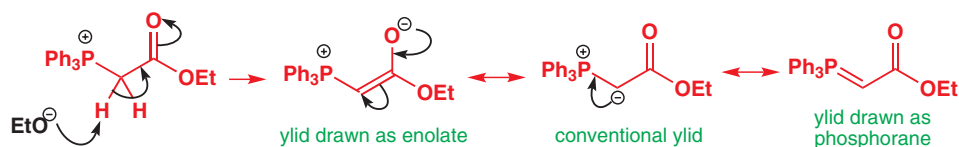
Conjugated Wittig reagents as specific enol equivalents

When the Wittig reaction was introduced (Chapter 11) we saw it simply as an alkene synthesis. Now if we look at one group of Wittig reagents, those derived from α -halo-carbonyl compounds, we can see that they behave as specific enol equivalents in making unsaturated carbonyl compounds.

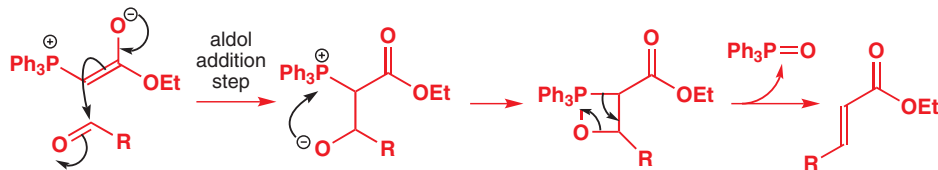


You notice that we have drawn the intermediate ylid as an enolate just to emphasize that it is an enolate derivative: it can also be represented either as the ylid or as an equivalent C=P 'phosphorane' structure. If we look at the details of this sort of Wittig reaction, we shall see that ylid formation is like enolate anion formation (indeed it is enolate anion formation). Only a weak base is needed as the enolate is stabilized by the Ph₃P⁺ group as well.

➡ We will return to ylids and the mechanism of the Wittig reaction in detail in Chapter 27.

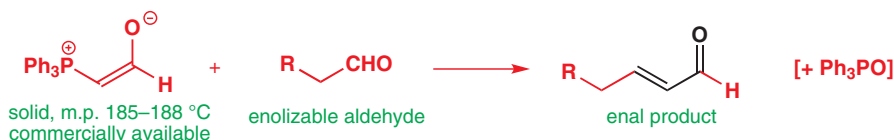


The first step of the Wittig reaction proper is just like an aldol reaction as it consists of an enolate attacking an electrophilic carbonyl compound. But, instead of forming an ‘aldol’ product, this adduct goes on to form an unsaturated carbonyl compound directly.



The final stages follow the mechanism of the Wittig reaction you met in Chapter 11: you can now see them as a special case of dehydration of an ‘aldol’ made favourable by the formation of a phosphine oxide and an unsaturated carbonyl compound.

The conjugated ylids derived from aldehydes, ketones, and esters are all sufficiently stable to be commercially available as the ylids—one of the few examples of specific enolate equivalents that you can actually buy. The ylid corresponding to the enolate of acetaldehyde is a solid, m.p. 185–188°C, that reacts well with other aldehydes, even if they are enolizable.

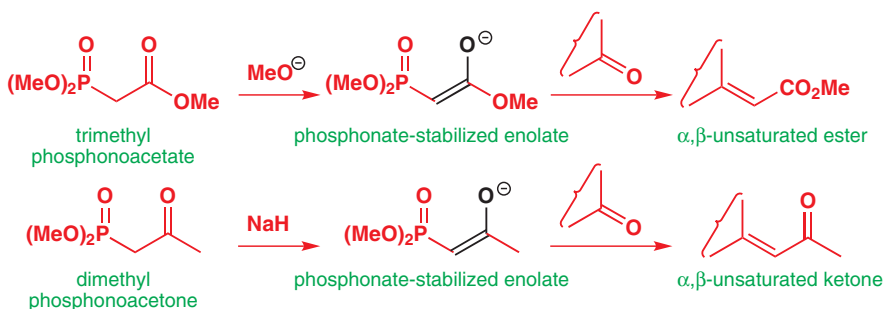


■ The alkene-forming reactions with (RO)₂P=O in the place of R₃P⁺ are known as Horner–Wadsworth–Emmons reactions. The Horner–Wadsworth–Emmons reaction can be used only to make conjugated alkenes.

■ The ‘brace’ device here is commonly used rather like ‘R’—it means that the rest of the molecule is unimportant to the reaction in question and could be anything.



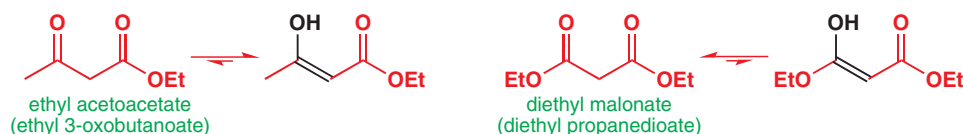
The stability of the phosphonium-stabilized enolates also means that, although they react well with aldehydes, their reactions with ketones are often poor, and it is better in these cases to use phosphonate-stabilized enolates. Being anionic, rather than neutral, these enolates are more nucleophilic. If an ester enolate equivalent is being used, the best base is the alkoxide ion belonging to the ester; with a ketone enolate equivalent, use sodium hydride or an alkoxide.



These last reagents, where the anion is stabilized both by the adjacent carbonyl group (as an enolate) and by the adjacent P=O group, are just one of many examples of enolate anions stabilized by two electron-withdrawing groups. The most important members of this class, enolates of 1,3-dicarbonyl compounds, are the subject of the next section.

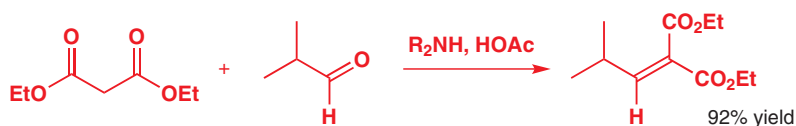
Specific enol equivalents from 1,3-dicarbonyl compounds

Although these are the oldest of the specific enol equivalents, they are still widely used because they need no special conditions—no low temperatures or strictly anhydrous solvents. The two most important are derived from malonic acid and ethyl acetoacetate.

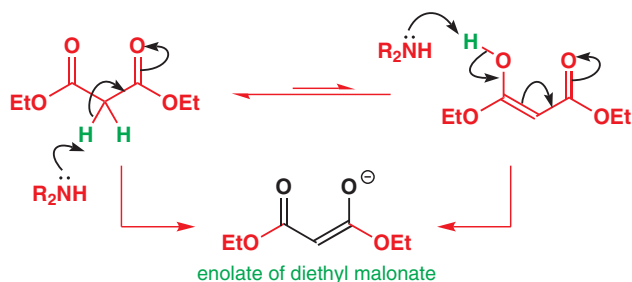


Ethyl acetoacetate is partly enolized under normal conditions. So, you might ask, why doesn't it immediately react with itself by the aldol reaction? There are two aspects to the answer. First, the enol is very stable (see Chapter 20 for a full discussion) and, second, the carbonyl groups in the unenolized fraction of the sample are poorly electrophilic ester and ketone groups. The second carbonyl group of the enol is not electrophilic because of conjugation. When a normal carbonyl compound is treated with catalytic acid or base, we have a small proportion of reactive enol or enolate in the presence of large amounts of unenolized electrophile. Aldol reaction (self-condensation) occurs. With 1,3-dicarbonyl compounds we have a small proportion of not particularly reactive unenolized compound in the presence of large amounts of stable (and hence unreactive) enol. No aldol occurs.

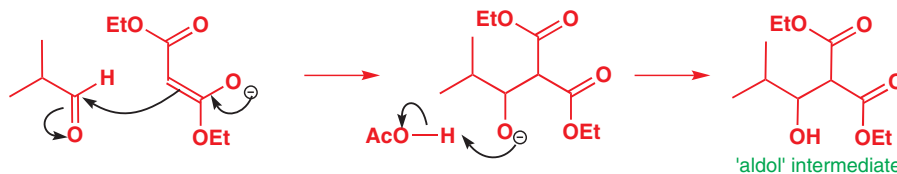
If we want a **crossed aldol reaction** with a 1,3-dicarbonyl compound, we simply add a second, electrophilic carbonyl compound such as an aldehyde, along with a weak acid or base. Often a mixture of a secondary amine and a carboxylic acid is used.



Reaction no doubt occurs via the enolate ion generated by the amine while the carboxylic acid buffers the solution, neutralizing the product and preventing enolization of the aldehyde. The amine (pK_a $R_2NH_2^+$ about 10) is a strong enough base to form a significant concentration of enolate from the 1,3-dicarbonyl compound (pK_a about 13) but not strong enough to form the enolate from the aldehyde (pK_a about 20). The formation of the enolate can be drawn from either tautomer of the malonate.

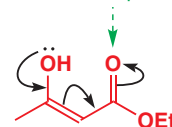


Now the enolate ion can attack the aldehyde in the usual way, and the buffer action of the acid produces the aldol product in the reaction mixture.



There is still one proton between the two carbonyl groups so enolate anion formation is again easy and dehydration follows to give the unsaturated product.

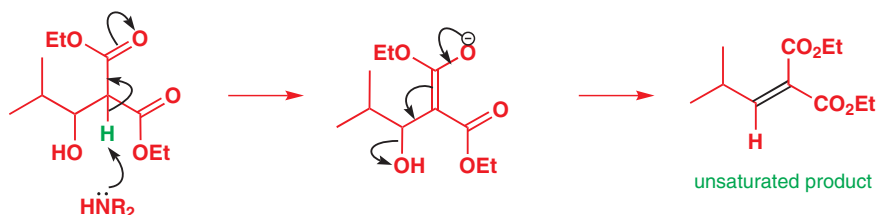
electrons are fed into this carbonyl group, making it less electrophilic



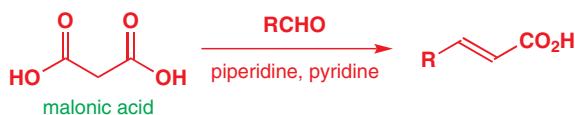
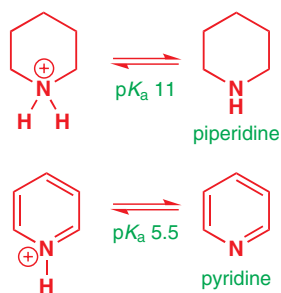
■ The aldol condensation of 1,3-dicarbonyl compounds under these conditions is sometimes called the **Knoevenagel reaction** after its nineteenth century inventor.

➡ *Tautomers* are isomers related to one another by *tautomerism*: see Chapter 20, p. 451.

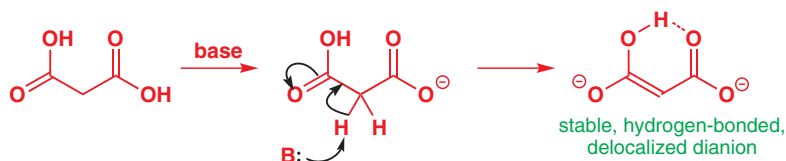
🔍 Interactive mechanism for the Knoevenagel reaction



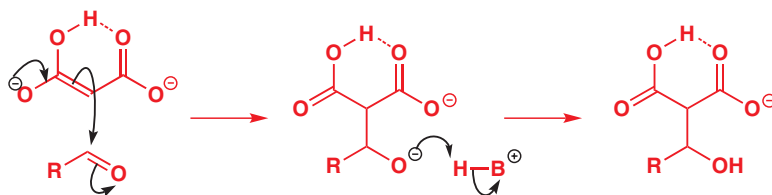
You may not want a product with both ester groups present, and we discussed in Chapter 25 how one of two 1,3-related ester groups may be removed by hydrolysis and decarboxylation. There is a simpler route with the aldol reaction. If, instead of the malonate diester, malonic acid is used, the decarboxylation occurs spontaneously during the reaction. The catalysts this time are usually a more basic mixture of piperidine and pyridine.



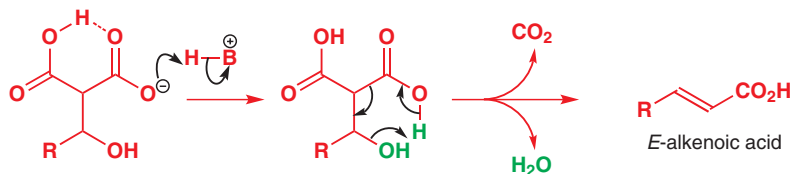
The reaction presumably uses the enolate anion of the monocarboxylate anion of malonic acid. Although this enolate is a dianion, its extensive delocalization and the intramolecular hydrogen bond make it really quite stable.



Next comes the aldol step. The dianion attacks the aldehyde, and after proton exchange the aldol is formed (still as the monocarboxylate in this basic solution).



Finally comes the decarboxylation step, which can occur through a cyclic mechanism (compare the decarboxylation mechanisms in Chapter 25). The decarboxylation could give either an *E* or a *Z* double bond depending on which acid group is lost as CO_2 , but the transition state leading to the more stable *E* product must be lower in energy since the product has *E* geometry.

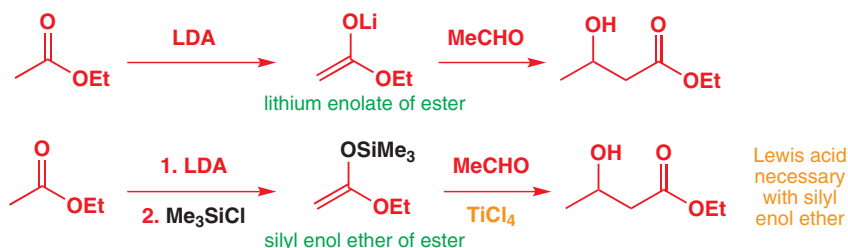


In the first part of this chapter we have looked at general solutions to the problem of controlling crossed aldol reactions. We'll now turn to the detailed ways those solutions are used with different classes of enolizable compounds.

How to control aldol reactions of esters

Among the enolates of carboxylic acid derivatives, esters are the most widely used. Ester enolates cannot be used as such in crossed aldols with aldehydes because the aldehyde is both more enolizable and more electrophilic than the ester. It will just condense with itself and ignore the ester. The same is true for ketones. A specific enol equivalent for the ester will therefore be needed for a successful aldol reaction of an ester enolate.

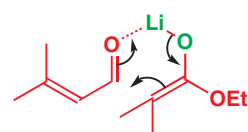
Fortunately, because this is a classic problem, many solutions are available. You can use the lithium enolate or the silyl enol ether, usually made best via the lithium enolate.



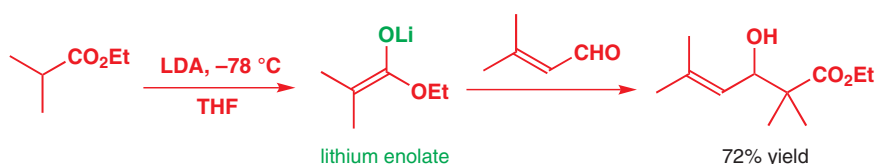
■ We have already discussed the special examples of malonate and phosphonoacetate esters above. Now we need to consider ester enolates more generally.

■ Forgive the reminder that a Lewis acid is necessary with silyl enol ethers.

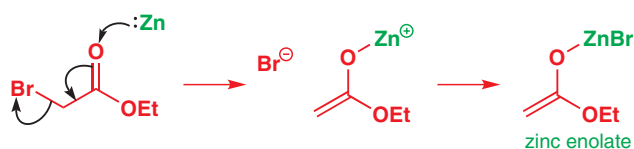
A good example is the first step in a synthesis of the natural product himalchene by Oppolzer and Snowden. Even though the ester and the aldehyde are both crowded with substituents, the aldol reaction works well with the lithium enolate of the ester. The cyclic mechanism ensures that the enolate adds directly to the carbonyl group of the aldehyde and not in a conjugate (Michael) fashion.



cyclic mechanism for ester aldol reaction

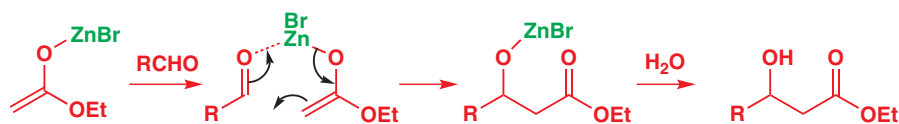


Zinc enolates, made from the bromoesters, are a good alternative to lithium enolates of esters. The mechanism for zinc enolate formation should remind you of the formation of a Grignard reagent.



There is no danger of self-condensation with zinc enolates as they do not react with esters. But they do react cleanly with aldehydes and ketones to give aldols on work-up. You will appreciate that the use of zinc enolates is therefore special to esters: you cannot make a zinc enolate from a 2-bromoaldehyde or an α -bromoketone as then you *would* get self-condensation.

■ Aldol reactions of zinc enolates formed in this way are known as Reformatsky reactions.



■ The dehydration product from this aldol product is best made directly by one of the Wittig variants we discussed earlier (p. 628). The same bromoester is, of course, the starting material for the ylid synthesis.

• Ester enolate equivalents

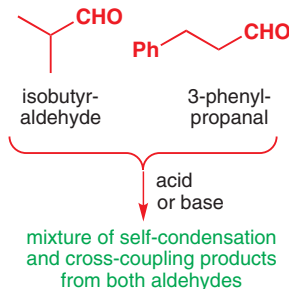
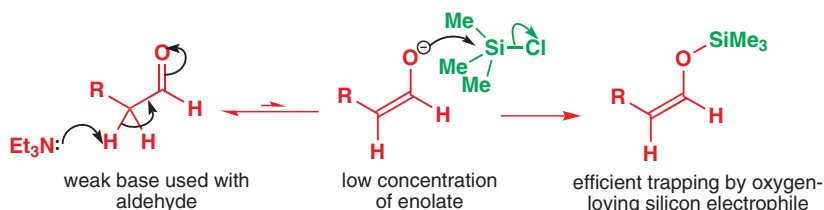
For aldol reactions with an ester enolate equivalent, use:

- lithium enolates *or*
- silyl enol ethers *or*
- zinc enolates.



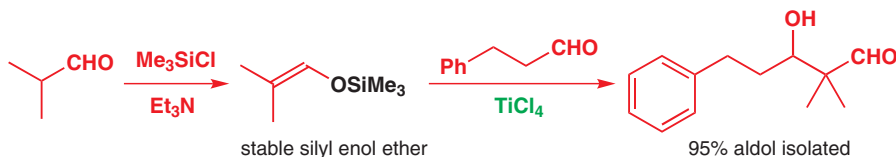
How to control aldol reactions of aldehydes

Aldehydes enolize very readily but also self-condense rather easily. Lithium enolates of aldehydes can't be made cleanly because the self-condensation reaction happens even at -78°C and is as fast as the enolization by LDA. Silyl enol ethers are a much better choice. They clearly must not be made via the lithium enolate, and amine bases are usually used. As each molecule of enolate is produced in the equilibrium, it is efficiently trapped by the silylating agent.

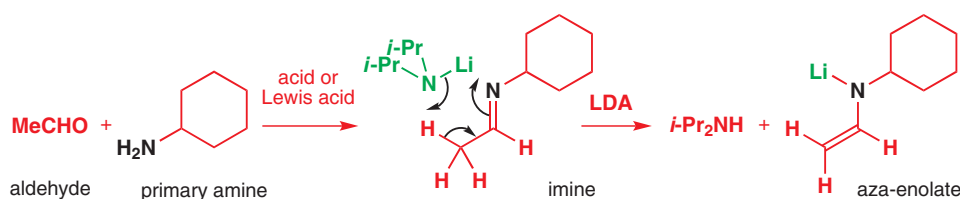


These silyl enol ethers are probably the best way of carrying out crossed aldol reactions with an aldehyde as the nucleophilic (enol or enolate) partner. An example is the reaction of the enol of the not very enolizable isobutyraldehyde with the very enolizable 3-phenylpropanal. Mixing the two aldehydes and adding base would of course lead to an orgy of self-condensation and cross-couplings.

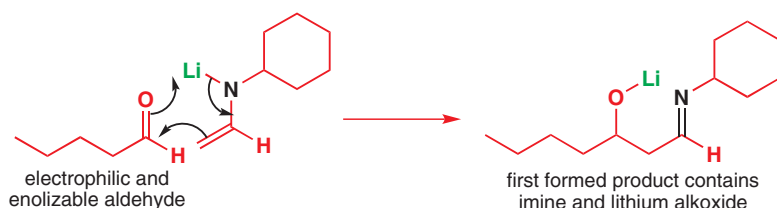
Preliminary formation of the silyl enol ether from either aldehyde, *in the absence of the other*, would be trouble-free as Me_3SiCl captures the enolate faster than self-condensation occurs. Here we need the silyl enol ether from isobutyraldehyde. The other aldehyde is now added along with the necessary Lewis acid, here TiCl_4 . The mechanism described on p. 627 gives the aldol after work-up in an excellent 95% yield. No more than 5% of other reactions can have occurred.



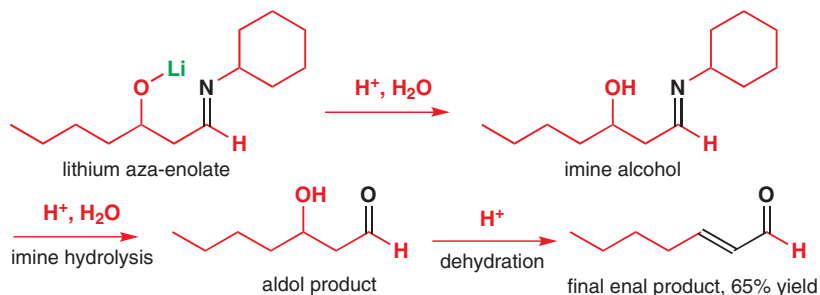
Other useful specific enol equivalents of aldehydes and ketones are enamines and aza-enolates, which you saw in use in alkylation reactions in Chapter 25. Aza-enolates—the lithium enolates of imines—derived from aldehydes are also useful in aldol reactions. Cyclohexylamine gives a reasonably stable imine even with acetaldehyde and this can be isolated and lithiated with LDA to give the aza-enolate. The mechanism is similar to the formation of lithium enolates and the lithium atom binds the nitrogen atom of the aza-enolate, just as it binds the oxygen atom of an enolate.



The aza-enolate reacts cleanly with other aldehydes or ketones to give aldol products. Even the most challenging of cross-couplings—attack on another similar enolizable aldehyde—occurs in good yield.



The initial product is a new imine, which is easily hydrolysed during acidic aqueous work-up. The alkoxide is protonated, the imine hydrolysed, and finally the aldol is dehydrated to give the enal—65% overall yield in this case.



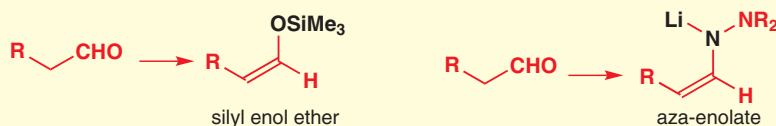
The key to the success of the aza-enolates is that the imine is first formed from the aldehyde with the primary amine, a relatively weak base, and under these conditions imine formation is faster than self-condensation. Only after the imine is formed is LDA added when self-condensation cannot occur simply because no aldehyde is left.

Except in certain cases (and you will meet some of these in Chapter 41) enamines are not generally used in aldol condensations, partly because they are not reactive enough, but mainly because they are too much in equilibrium with the carbonyl compound itself and exchange would lead to self-condensation and the wrong cross-couplings. You will see later that enamines come into their own when we want to acylate enols with the much more reactive acid chlorides.

● Aldehyde enolate equivalents

For crossed aldol reactions with an aldehyde as the enol partner, use:

- silyl enol ethers or
- aza-enolates.



For acylation of aldehyde enolates (see later), use silyl enol ethers or enamines.

■ Imines are susceptible to hydrolysis and they are best not stored but used at once. To understand these reactions fully you should ensure you are familiar with the mechanisms of imine formation and hydrolysis from Chapter 11.

How to control aldol reactions of ketones

The enolization of ketones, unless they are symmetrical, poses a special problem. Not only do we need to prevent them self-condensing (although this is less of a problem than with aldehydes), but we also need to control which side of the carbonyl group the ketone enolizes. In this section we shall introduce aldol reactions with unsymmetrical ketones where one of two possible enols or enolates must be made.

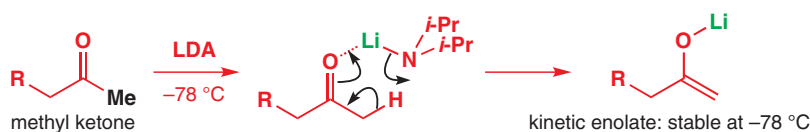
Making the less substituted enolate equivalent: kinetic enolates

Treatment of methyl ketones with LDA usually gives only the lithium enolate on the methyl side. This is the enolate that forms the fastest and is therefore known as the kinetic enolate. It is formed faster *because*:

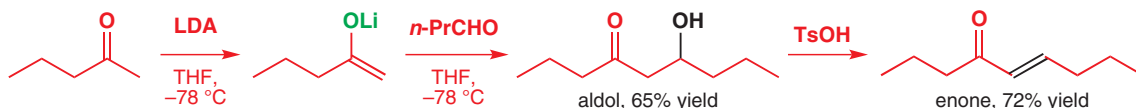
- the protons on the methyl group are more acidic
- there are three of them as against two on the other side, and
- there is steric hindrance to attack by LDA on the other side of the carbonyl group.

➡ Kinetic and thermodynamic enolates were introduced in Chapter 25, p. 601.

Gilbert Stork was born in Brussels and became an assistant professor of chemistry at Harvard in 1948. From 1953, Stork was at Columbia University in New York. He pioneered new synthetic methods, among them many involving enolates and enamines.

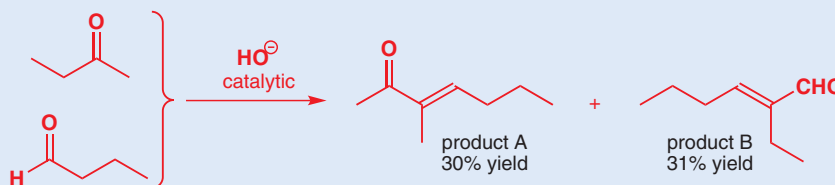


A simple example from the first report of this reaction by Gilbert Stork and his group in 1974 is the condensation of pentan-2-one with butanal to give the aldol and then the enone oct-4-en-3-one by acid-catalysed dehydration. The yields may seem disappointing, but this was the first time anyone had carried out a crossed aldol reaction like this with an unsymmetrical ketone and an enolizable aldehyde and got just one aldol product in any reasonable yield at all.

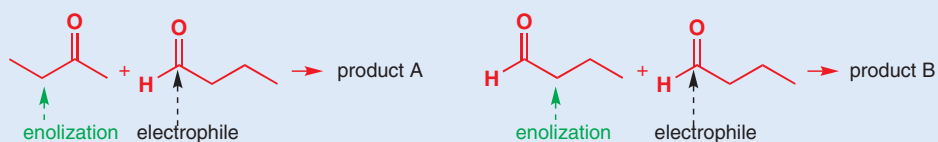


An uncontrolled ketone aldol

A more typical result from the days before specific enol condensation between butanone and butanal with equivalents had been invented is this attempted crossed catalytic base. Two products were isolated in low yield.



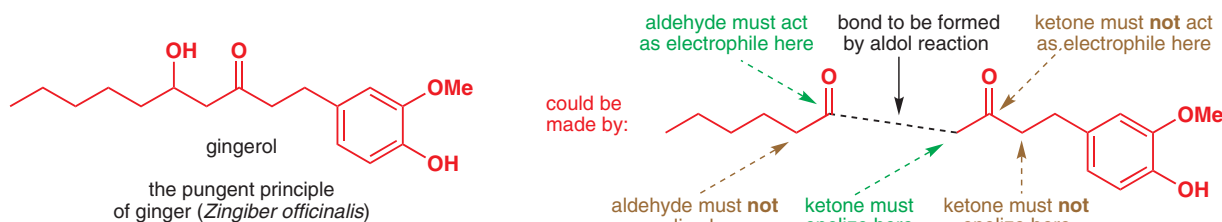
Product A is from the enolate of the more substituted side of the ketone reacting with the aldehyde, and product B is just the self-condensation product from the aldehyde.



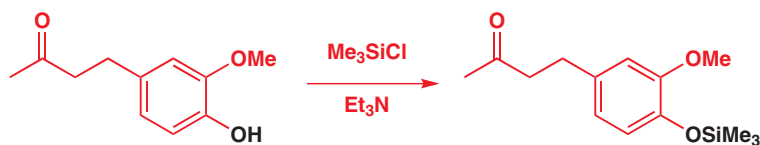
These kinetic lithium enolates are stable in THF at -78°C for a short time but can be preserved at room temperature in the form of their silyl ethers.



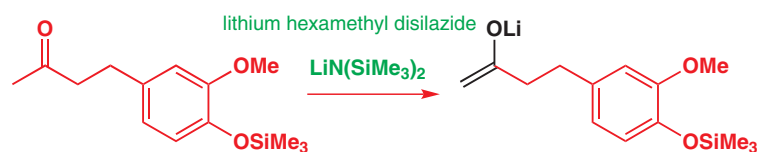
Aldol reactions can be carried out with either the lithium enolate or the silyl enol ether. As an example we shall use the synthesis of a component of the flavour of ginger. The hotness of ginger comes from ‘gingerol’—the ‘pungent principle’ of ginger. Gingerol is a 3-hydroxyketone, so we might consider using an aldol reaction to make it. We shall need the enol (or enolate) on the methyl side of an unsymmetrical ketone to react with a simple aldehyde (pentanal) as the electrophilic partner in the aldol reaction. Pentanal is an enolizable aldehyde, so we must stop it enolizing. The diagram summarizes the proposed aldol reaction.



We might consider using the lithium enolate or the silyl enol ether. As we need the kinetic enolate (the enolate formed on the less substituted side of the ketone), we shall be using the lithium enolate to make the silyl enol ether, so it would make sense to try that first. There is another problem too. The ketone has a free OH group on the far side of the ring that will interfere with the reaction. We must protect that first as an ordinary silyl ether (not a silyl *enol* ether).

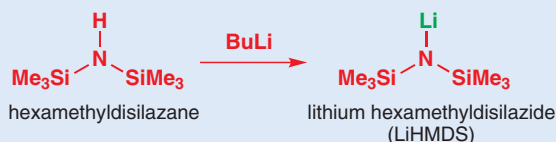


Now we can make the kinetic lithium enolate with a hindered lithium amide base. In fact, the one chosen here was even more hindered than LDA as it has two Me₃Si groups on the nitrogen atom.



Lithium hexamethyldisilazide

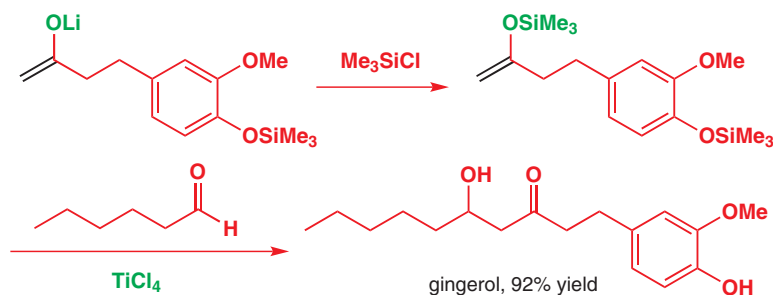
Lithium hexamethyldisilazide (LiHMDs) is a little more hindered than LDA and a little less basic. It is made by deprotonating hexamethyldisilazane with BuLi.



An aldol reaction with this lithium enolate on pentanal was successful and the protecting group (the silyl ether) was conveniently hydrolysed during work-up to give gingerol itself. However, the yield was only 57%. When the silyl enol ether was used with TiCl₄ as the Lewis

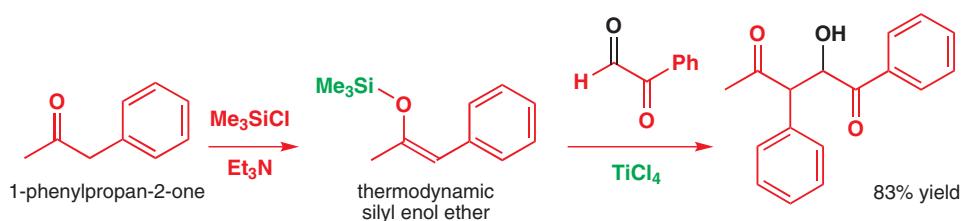
Teruaki Mukaiyama, of the Science University of Tokyo (and formerly of the Tokyo Institute of Technology and the University of Tokyo), was one of the foremost Japanese chemists of his generation, whose work has had a significant impact on the development of the aldol reaction and on other areas of organic synthesis.

acid catalyst, the yield jumped to 92%. This is one of the many successful uses of this style of aldol reaction by Mukaiyama, the inventor of the method.



Making the more substituted enolate equivalent: thermodynamic enolates

Being an alkene, an enol or enolate is more stable if it has more substituents. So the way to make the more substituted enolate equivalent is to make it under conditions where the two enolates can interconvert: equilibration will give the more stable form. You have seen in Chapter 25 (p. 599) how the silyl enol ether on the more substituted side of a ketone can be made by treating the ketone with Me_3SiCl and a weak base, but these thermodynamic silyl enol ethers have been little used in aldol reactions. One successful example is the thermodynamic silyl enol ether of 1-phenylpropan-2-one: enolization on the conjugated side is overwhelmingly favoured thermodynamically. The aldol reaction with a 2-ketoaldehyde goes exclusively for the more reactive aldehyde group.

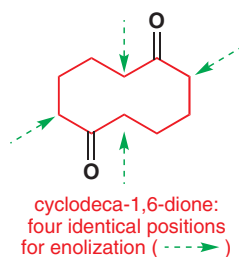


This concludes our general survey of specific enolates in the aldol reaction. Later you will see many of the same reagents used in acylation at carbon. We are left with some reactions that are particularly easy to do.

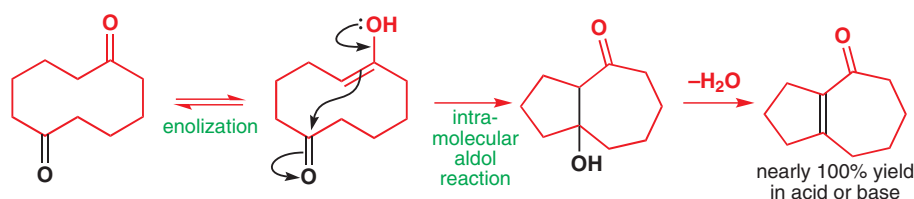
Intramolecular aldol reactions

Now for something easy. When an aldol reaction can form a five- or six-membered ring, you need no longer worry about specific enols or anything like that. Equilibrium methods with weak acids or bases are quite enough to give the cyclic product by an intramolecular aldol reaction because intramolecular reactions are faster than intermolecular ones. We shall illustrate intramolecular reactions by looking at the cyclization of a series of diketones of increasing complexity, starting with one that can form four equivalent enols: cyclodeca-1,6-dione.

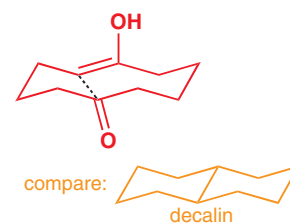
It doesn't matter where enolization occurs because the same enol is formed. And once the enol is formed, there is only one thing it can reasonably do: attack the other ketone to form a stable five-membered ring. It also gives a reasonably stable seven-membered ring, but that is by the way. In weak acid or base, only a small proportion of carbonyl groups will be enolized, so the chance of two being in the same molecule is very low. No intermolecular condensation is found and the yield of the bicyclic enone from the intramolecular reaction is almost 100% (96% with Na_2CO_3).



➡ Ring size and stability were discussed in Chapter 16.

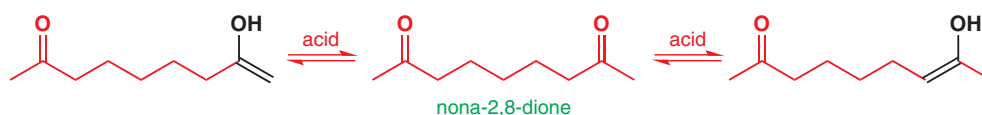


This may look like a long stretch for the enol to reach across the ten-membered ring to reach the other ketone, but the conformational drawing in the margin shows just how close they can be. You should compare this conformation with that of a decalin (Chapter 16).

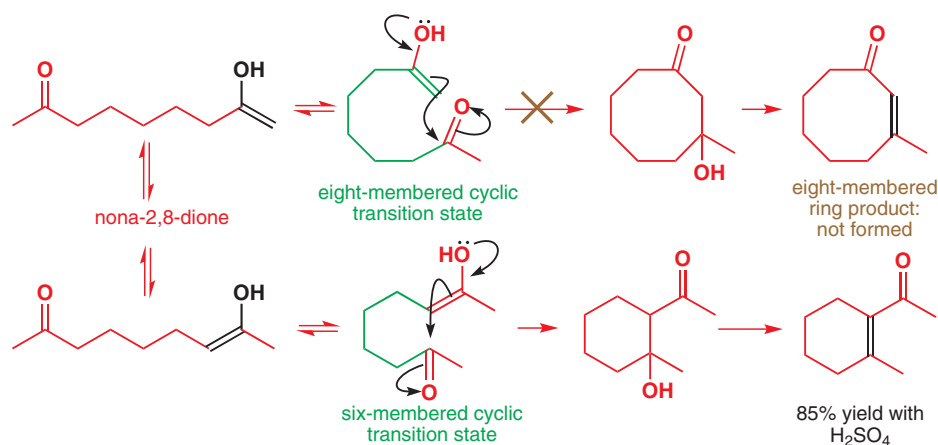


- Intramolecular reactions giving five- or six-membered rings are preferred to those giving strained three- or four-membered rings on the one hand or medium rings (eight- to thirteen-membered) on the other.

Acid-catalysed cyclization of the symmetrical diketone nona-2,8-dione could give two enols.

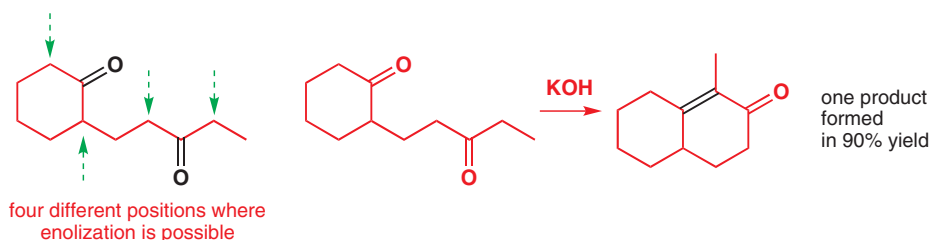


One enol can cyclize through an eight-membered cyclic transition state and the other through a six-membered one. In each case the product would first be formed as an aldol but would dehydrate to the cyclic enone having the same ring size as the transition state. In practice, only the less strained six-membered ring is formed and the enone can be isolated in 85% yield.

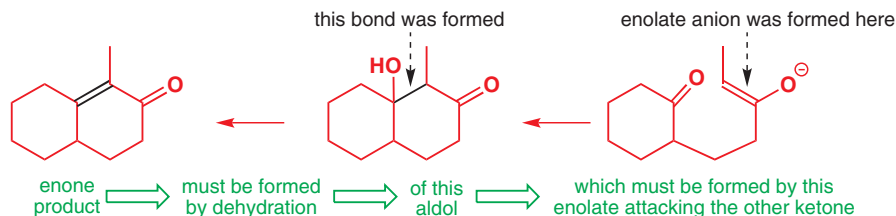


Interactive mechanism for intramolecular aldol reactions

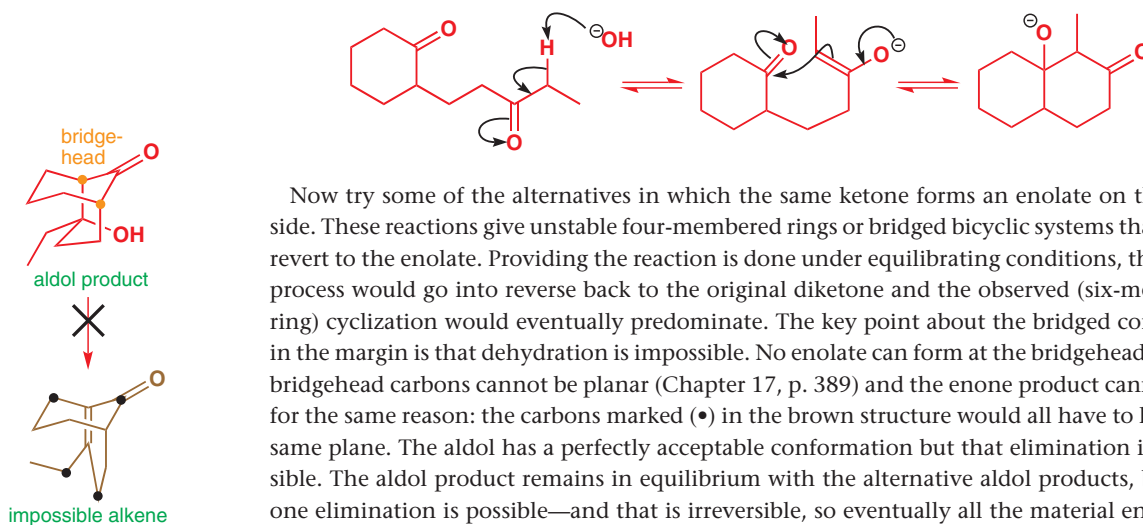
Most diketones lack symmetry, and will potentially have four different sites for enolization. Consider what might happen when this diketone is treated with KOH. There are four different places where an enolate anion might be formed as there are four different carbon atoms. There are also two different electrophilic carbonyl groups so that there are many possibilities for inter- and intramolecular condensation. Yet only one product is formed, in 90% yield.



We can deduce the mechanism of the reaction simply from the structure of the product by working backwards. The double bond is formed from an aldol whose structure we can predict and hence we can see which enolate anion was formed and which ketone acted as the electrophilic partner.



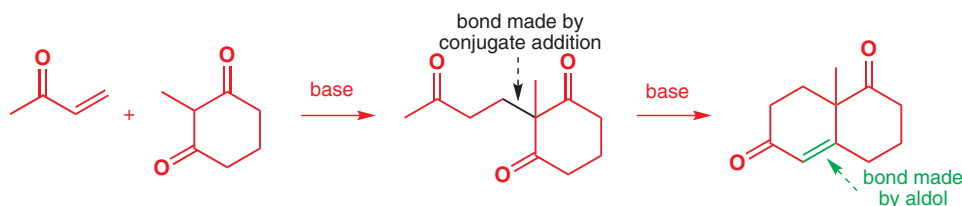
Must we argue that this one enolate is more easily formed than the other three? No, of course not. There is little difference between all four enolates and almost no difference between the three enolates from CH_2 groups. We *can* argue that this is the only aldol reaction that leads to a stable conjugated enone in a stable six-membered ring. This must be the mechanism; the others are just too slow to compete. Protonation and dehydration follow as usual.



Now try some of the alternatives in which the same ketone forms an enolate on the other side. These reactions give unstable four-membered rings or bridged bicyclic systems that would revert to the enolate. Providing the reaction is done under equilibrating conditions, the whole process would go into reverse back to the original diketone and the observed (six-membered ring) cyclization would eventually predominate. The key point about the bridged compound in the margin is that dehydration is impossible. No enolate can form at the bridgehead because bridgehead carbons cannot be planar (Chapter 17, p. 389) and the enone product cannot exist for the same reason: the carbons marked (•) in the brown structure would all have to lie in the same plane. The aldol has a perfectly acceptable conformation but that elimination is impossible. The aldol product remains in equilibrium with the alternative aldol products, but only one elimination is possible—and that is irreversible, so eventually all the material ends up as the one enone.

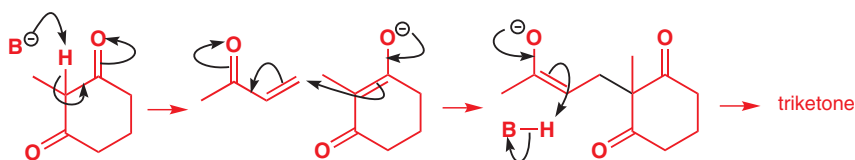
The Robinson annelation

One of the most important applications of the intramolecular aldol reaction is a ring synthesis (annellation or annulation) that takes place in two steps, both involving enols. The compound made by Robinson in the first example is a bicyclic diketone that contains the basic structure of rings A and B of the steroids. The bonds made in the two steps are marked.

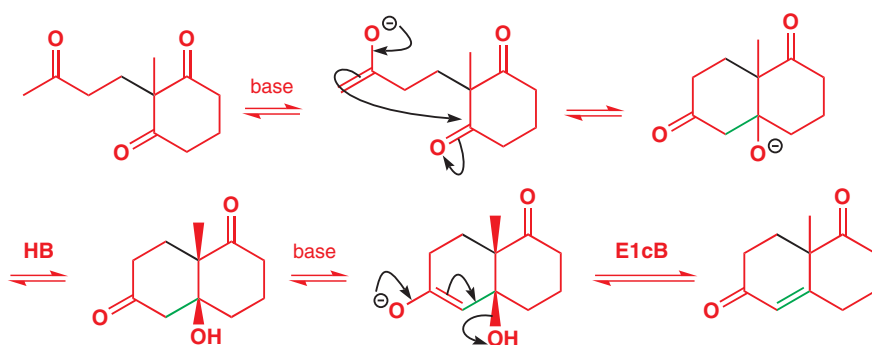


Robert Robinson (1886–1975) was a British chemist who won the Nobel Prize in 1947 for his work on the synthesis of alkaloids.

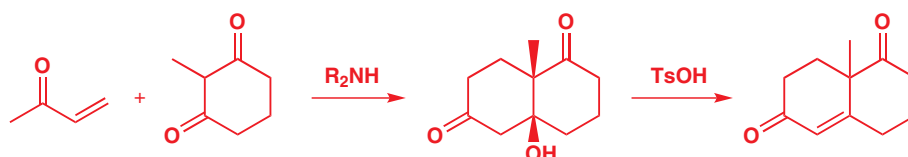
Only a weak base is needed to form the stable enolate of the 1,3-diketone and this does conjugate addition onto the enone (Chapter 25). The intermediate triketone may be isolated but often isn't.



The second stage starts with the intramolecular aldol reaction. You should be able to see that the alternatives to a six-membered ring are a four-membered ring and bridged products. The hydroxy-ketone, which happens to have the *cis* stereochemistry, can also be isolated but eliminates by the E1cB mechanism to complete the aldol sequence.



Other ways to carry out this same reaction are to use a secondary amine as the weak base. This gives an excellent yield of the hydroxyketone that can be converted into the enone with acid.

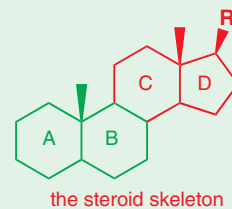


In this sequence the new ring is built onto the side of an old ring but this is not necessary. Any combination of an easily enolizable compound and an enone may give a Robinson annulation product. A simple example combines a non-enolizable enone with ethyl acetoacetate to give an excellent yield of a cyclohexenone. As these compounds are so robust a stronger base can be used.



The Darzens reaction

Tandem reactions, in which a second enolate reaction follows on from the first, can allow us to make cyclopropanes (see Chapter 25, p. 586), by conjugate addition followed by C-alkylation, or epoxides, by aldol addition followed by O-alkylation. This epoxide was used in the synthesis of the drug Darusentan.

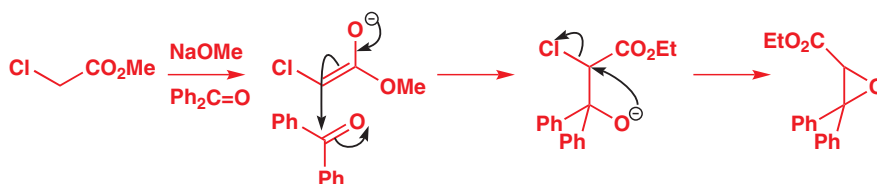


the steroid skeleton

There is more on steroids in Chapter 42.

Interactive mechanism for Robinson annulation

➔ You will discover in Chapter 41 that using the natural amino acid proline as the amine favours a single enantiomer of this product.

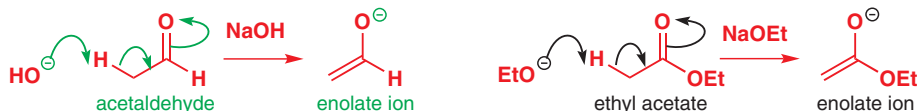


The formation of epoxides in this way complements their formation from alkenes with *m*-CPBA because it involves construction of a C–C bond. Epoxide formation from α -halogenated carbonyl compounds is known as the **Darzens reaction**.

Acylation at carbon

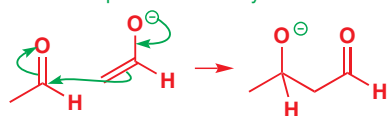
Introduction: the Claisen ester condensation and the aldol reaction compared

We began this chapter with the treatment of acetaldehyde with base. This led initially to the formation of an enolate anion and then to the aldol reaction. We are going to start this section by looking at what happens if you just treat ethyl acetate with base. To start with, there is hardly any difference. We shall use ethoxide as base rather than hydroxide as hydroxide would hydrolyse the ester, but otherwise the first steps are very similar. Here they are, side by side.

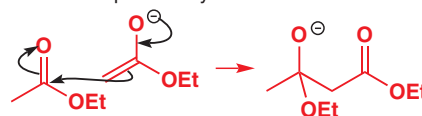


The next step in both cases is nucleophilic attack by the enolate ion on unenolized carbonyl compound. The concentration of enolate is low and each enolate ion is surrounded by unenolized aldehyde or ester molecules, so this reaction is to be expected. Here is that step, again shown for both aldehyde and ester.

the aldol step with acetaldehyde

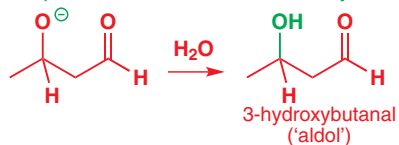


the 'aldol' step with ethyl acetate

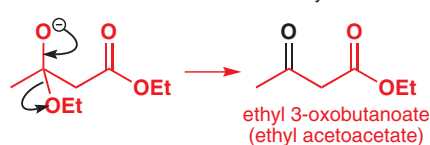


Only now does something different happen. The aldehyde dimer simply captures a proton from the solvent to give an aldol product. The 'aldol' from the ester (not, in fact, an aldol at all) has a leaving group, EtO^- , instead of a hydrogen atom and is actually the tetrahedral intermediate in a nucleophilic substitution at the carbonyl group. Compare the two different steps again.

completion of the aldol with acetaldehyde

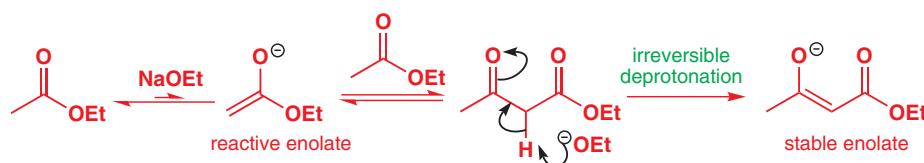


the Claisen condensation with ethyl acetate



Even though the last step is different, the two products are quite similar. Both are dimers of the original two-carbon chain and both have carbonyl groups at the end of the chain and oxygen substituents at position three. The two reactions obviously belong to the same family but are usually given different names. The ester reaction is sometimes known as the **Claisen ester condensation** and sometimes as the **Claisen–Schmidt reaction**. More important than remembering the name is being familiar with the reaction and its mechanism.

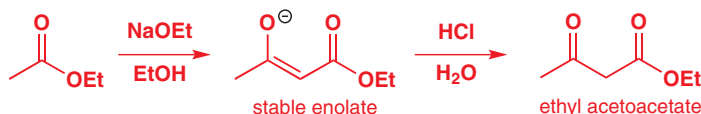
This is another of those reactions where the base is not strong enough to transform the ester entirely into the enolate. Only a small equilibrium concentration is produced, which reacts with the ester electrophile. The by-product from the reaction is ethoxide ion and so it looks at first sight as though we get our catalyst back again—the aldol, if you remember, is catalytic in base. But not the Claisen reaction. The second step of the reaction is also really an equilibrium, and the reaction works only because the product can be irreversibly deprotonated by the ethoxide by-product, consuming ethoxide in the process. You recall that the aldol reaction often works best when there is an extra driving force to push it across—dehydration to an enone, for example. Similarly, the ester dimerization works best when the product reacts with the ethoxide ion to give a stable enolate ion.



Interactive mechanism for Claisen ester condensation

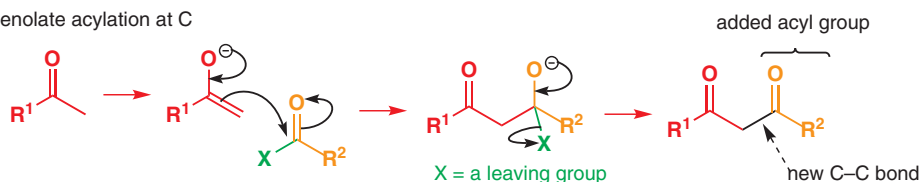
The point is that the base used, ethoxide ion EtO^- , is too weak (EtOH has a pK_a of about 16) to remove the proton completely from ethyl acetate (pK_a about 25), but is strong enough to remove a proton from the acetoacetate product (pK_a about 10). Under the conditions of the reaction, a small amount of the enolate of ethyl acetate is produced—just enough to let the reaction happen—but the product is completely converted into its enolate. The neutral product, ethyl acetoacetate itself, is formed on acidic work-up.

the complete Claisen ester condensation



The final product has been formed by the acylation at carbon of the enolate of an ester. This general process—acylation at carbon—is the subject of the second part of this chapter. It so happened in this case that the acylating agent was another molecule of the same ester, but the general process we shall consider is the acylation of enolates at carbon. We shall use a variety of enols, enolates, and specific enol equivalents and a variety of acylating agents, but the basic idea is that the enolate of one carbonyl compound will have an acyl group (here the R^2CO group in orange) added to the enolate carbon atom.

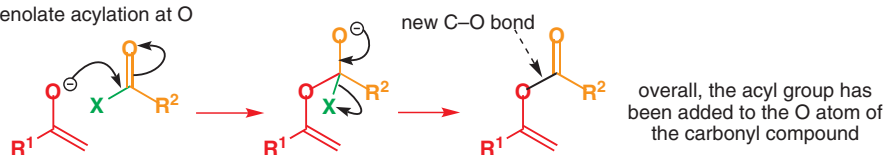
enolate acylation at C



Problems with acylation at carbon

The main problem with the acylation of enolates is that reaction tends to occur at oxygen rather than at carbon.

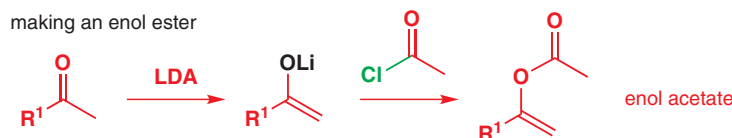
enolate acylation at O



overall, the acyl group has been added to the O atom of the carbonyl compound

You have seen reaction at oxygen before. Enolates react on oxygen with silicon electrophiles and we found the products, silyl enol ethers, useful in further reactions. Enol esters also have their uses—as precursors of lithium enolates, for example. You saw one being used like this on p. 454.

The product of acylation on oxygen is an **enol ester**. The tendency to attack through oxygen is most marked with reactive enolates and reactive acylating agents. The combination of a lithium enolate and an acid chloride, for example, is pretty certain to give an enol ester.



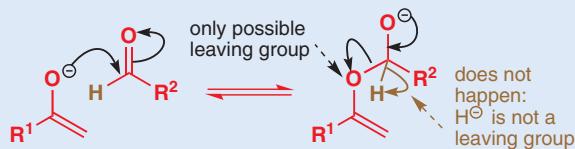
If we want acylation at carbon we must use *either*:

- less reactive specific enol equivalents, such as enamines or silyl enol ethers, with reactive acylating agents such as acid chlorides *or*
- reactive enols, such as the enolate anions themselves, with less reactive acylating agents such as esters.

We introduced this chapter with an example of the second type of reaction, and we shall continue with a more detailed consideration of the Claisen ester condensation and related reactions.

Reaction at oxygen—not a problem in the aldol reaction

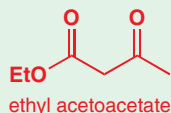
Earlier in this chapter, we mentioned no trouble with reaction at oxygen in the aldol reaction. This may now seem surprising, in view of what we have said about esters, as the electrophiles were aldehydes and ketones—not so very different from esters. We can resolve this by looking at what would happen if an aldehyde did attack an enolate on the oxygen atom.



The only plausible leaving group from the intermediate is the enolate oxygen: the reaction just reverses.

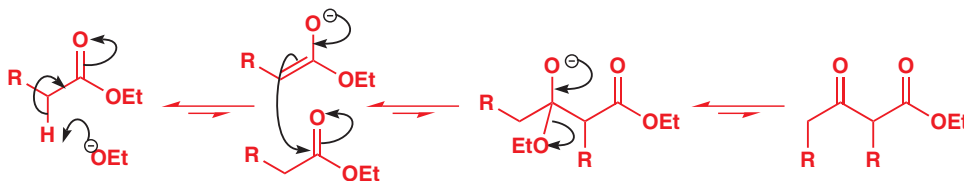
The Claisen ester condensation and other self-condensations

■ We have already considered reactions of ethyl acetoacetate: now you see how it is made.

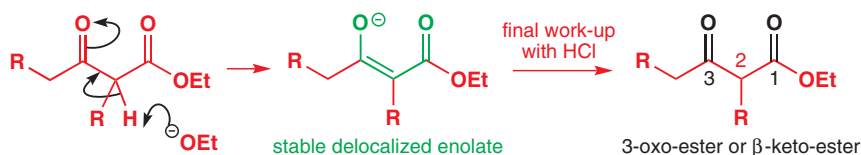


The self-condensation of ethyl acetate is the most famous example of the Claisen ester condensation and it works in good yield under convenient conditions. The product (ethyl acetoacetate) is commercially available for this very reason—and cheap too—so you are unlikely to want to do this particular example.

A more generally useful reaction is the self-condensation of simple substituted acetates $\text{RCH}_2\text{CO}_2\text{Et}$. These work well under the same conditions (EtO^- in EtOH). The enolate anion is formed first in low concentration and in equilibrium with the ester. It then carries out a nucleophilic attack on the more abundant unenolized ester molecules.



These steps are all unfavourable equilibria and, on their own, would give very little product. However, as we mentioned before, the reaction works because the equilibrium is driven over by the essentially irreversible formation of a stable, delocalized enolate from the product.

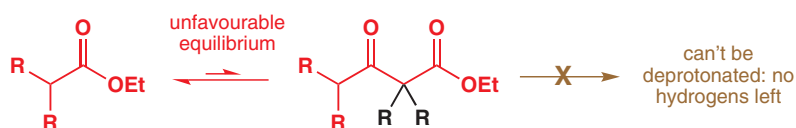


Finally, the reaction is worked up in acid and the β keto-ester product is formed. Notice that all products of Claisen ester condensations have a 1,3-dicarbonyl relationship. These compounds are useful in the preparation of specific enol equivalents and you have seen them in action in Chapters 20 and 25, and in this chapter.

➔ We shall discuss the significance of the 1,3-relationship in Chapter 28.

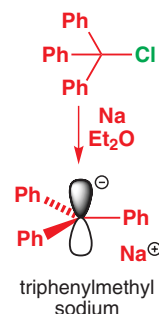
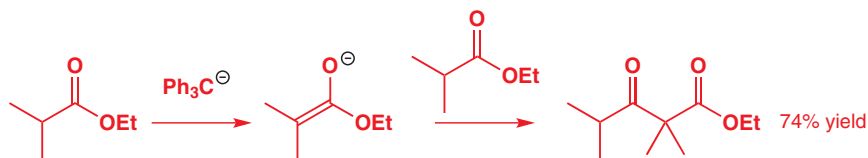
How do we know that deprotonation drives the reaction?

If the original ester has two substituents on the α carbon atom (C2 of the ester), the formation of the stable enolate of the product is no longer possible as there are no hydrogen atoms left to remove.



As you might expect, all the equilibria are now unfavourable, and this reaction does not go well under the normal equilibrating conditions (EtO^- in EtOH). It can be made to go in reasonable yield if a stronger base is used. Traditionally, triphenylmethyl sodium is chosen. This is made from Ph_3CCl and sodium metal, and is a very conjugated carbanion.

Triphenylmethyl carbanion is a strong enough base to convert an ester entirely into its enolate. Reaction of the enolate with a second molecule of ester then gives the keto-ester in good yield.



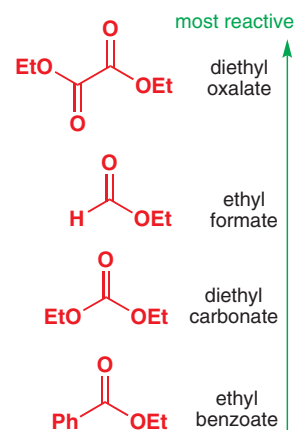
Crossed ester condensations

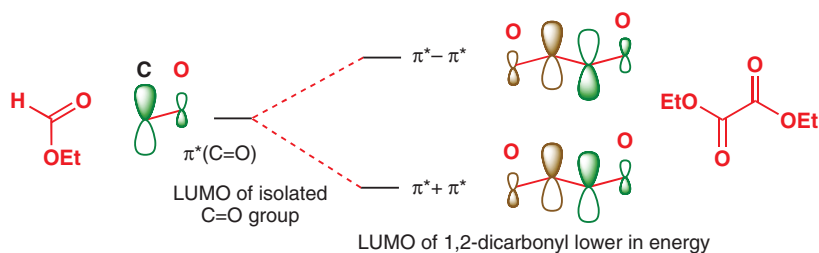
Much the same arguments apply here as applied in the crossed aldol reaction. We must be quite sure that we know which compound is going to act as the enol partner and which as the acylation partner.

Reactive esters that cannot enolize

There are several useful esters of this kind, of which the four in the margin are the most important. They cannot act as the enol partner, and the first three are more electrophilic than most esters, so they should acylate an ester enolate faster than the ester being enolized can.

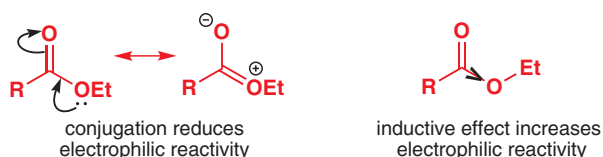
These four are arranged in order of reactivity towards nucleophiles, the most electrophilic at the top and the least electrophilic at the bottom. Oxalates are very reactive because each carbonyl group makes the other more electrophilic. The molecular LUMO is the *sum* of the two π^* orbitals and is lower in energy than either.



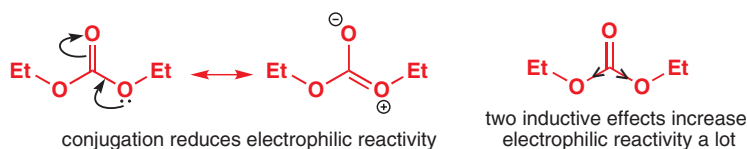


Formate esters look a bit like aldehydes but their ester character dominates. The hydrogen atom just makes them very electrophilic as they lack the σ conjugation (and steric hindrance) of simple esters.

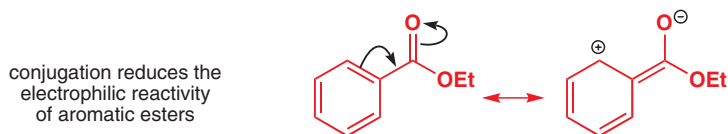
Carbonates are particularly useful as they introduce a CO_2R group on to an enolate. It is perhaps not immediately obvious why they are more electrophilic than simple esters. Normal esters are (slightly) less electrophilic than ketones because the deactivating lone pair donation by the oxygen atom is more important than the inductive effect of the electronegative oxygen atom.



The result is a small difference between two large effects. In carbonate esters there are two oxygen atoms on the same carbonyl group. Both can exert their full inductive effect but the lone pairs have to share the same π^* orbital. The balance is changed—the summed inductive effects win out—and carbonates are more electrophilic than ordinary esters.

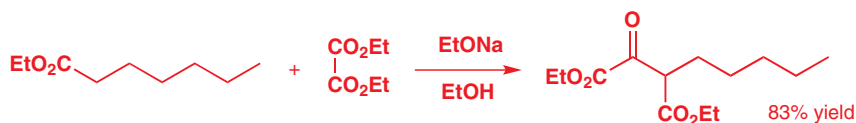


Finally, esters of aromatic acids cannot enolize but are less reactive than ordinary esters because of conjugation from the aromatic ring. These compounds may still be useful, as we shall see.



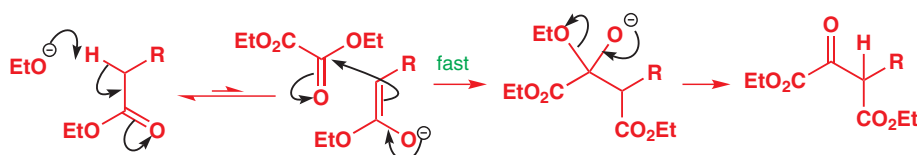
Crossed Claisen ester condensations between two different esters

To illustrate some Claisen reactions which are easy to do, we shall now give a few examples of crossed Claisen ester condensations between ordinary esters and the compounds we have just discussed. First, a reaction between a simple linear ester and diethyl oxalate performed under equilibrating conditions with ethoxide as the base. The weak base means a lower enolate concentration.

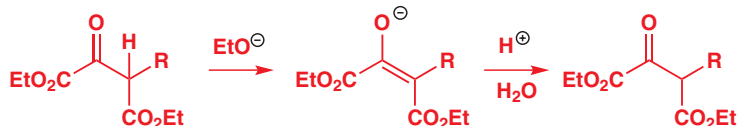


Only the simple ester can give an enolate, and the low concentration of this enolate reacts preferentially with the more electrophilic diethyl oxalate in a typical acylation

at carbon. No self-condensation of the simple ester occurs as the oxalate is much more electrophilic.



The product has an acidic hydrogen atom so it is immediately converted into a stable enolate, which is protonated on work-up in aqueous acid to give the tricarbonyl compound back again.



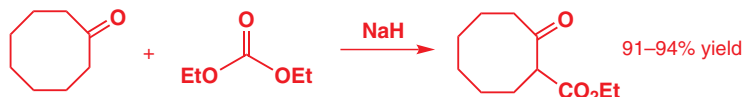
Another important example leads to the preparation of diethyl phenylmalonate. This compound cannot be made by 'alkylation' of diethyl malonate as aryl halides do not undergo nucleophilic substitution (Chapter 22).

A crossed Claisen ester condensation between very enolizable ethyl phenylacetate and unenolizable but electrophilic diethyl carbonate works very well indeed under equilibrating conditions.

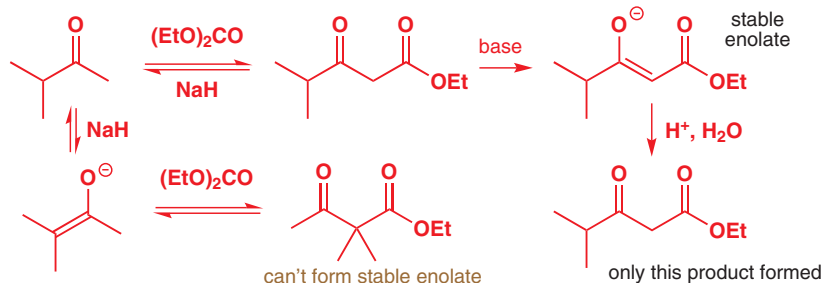


Claisen condensations between ketones and esters

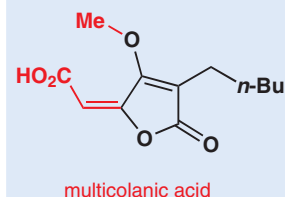
Claisen condensations are acylations that always involve esters as the electrophilic partner, but enolates of other carbonyl compounds—ketones, for example—may work equally well as the enol partner. In a reaction with a carbonate, only the ketone can enolize and the reactive carbonate ester is more electrophilic than another molecule of the ketone. A good example is this reaction of cyclooctanone. It does not matter which side of the carbonyl group enolizes—they are both the same.



Unsymmetrical ketones often give a single product, even without the use of a specific enol equivalent, as reaction usually occurs on the less substituted side. This is another consequence of the final enolization being the irreversible step. In this example, both possible products may form, but only one of them can form a stable enolate. Under the equilibrating conditions of the reaction, only the enolate is stable, and all the material ends up as the isomer shown.

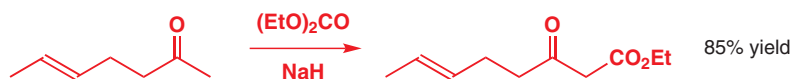


This compound was made because it was needed in a synthesis of multicolic acid, a metabolite of a penicillium mould. It is easy to see which atoms of the natural product (shown in black) were provided by the compound we have just made in a single easy step.

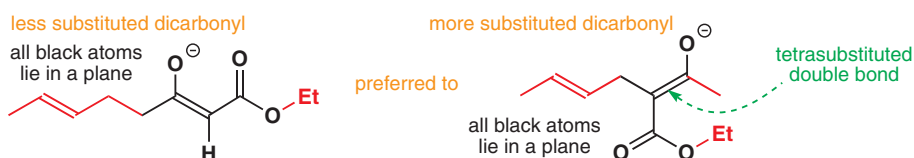


■ We're including carbonates as esters here: they are esters of carbonic acid.

Unsymmetrical ketones work well even when one side is a methyl group and the other a primary alkyl chain. This example gives an impressive yield and shows that, as expected, a remote alkene does not affect the reaction.



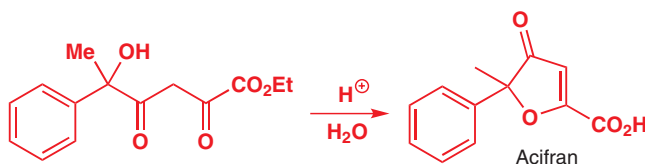
Even when both enolates can form, the less substituted dicarbonyl enolate is preferred because it constrains fewer groups to lie in the hindered plane of the tetrasubstituted enolate double bond.



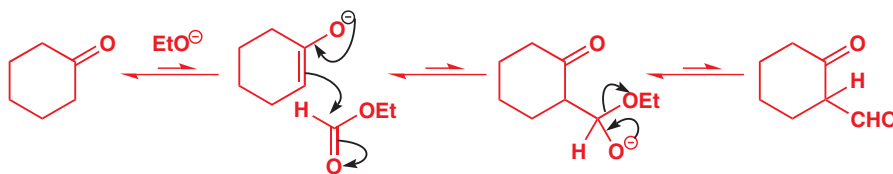
Diethyl oxalate also gives well-controlled condensations with ketones and we shall take the synthesis of a new drug as an example. One way to try to prevent heart disease is to reduce the amount of 'bad' lipoproteins in the blood. The drug Acifran does this, and a key step in its synthesis is the base-catalysed reaction between diethyl oxalate and a methyl ketone.



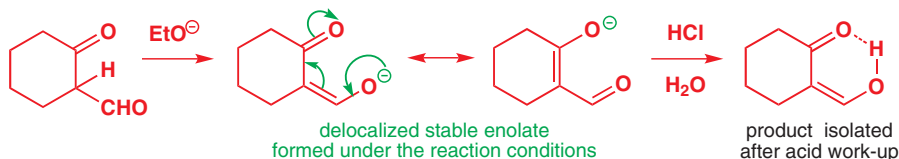
Notice that the hydroxyl group on the ketone does not interfere with the reaction. No doubt the first molecule of base removes the OH proton and the second molecule forms the enolate (the only possible enolate in either molecule). Fast condensation with highly electrophilic diethyl oxalate follows. The drug itself results from simple acid treatment of this product.



The other two unenolizable esters we mentioned on p. 643 undergo cross-condensations with ketones. Unlike formaldehyde, formate esters are well behaved—no special method is necessary to correspond with the Mannich reaction in aldol chemistry. Here is what happens with cyclohexanone.



The product aldehyde is not at risk from nucleophilic attack, as it appears to be, because it immediately enolizes in base. On work-up, the product is formed as a stable enol with an intramolecular hydrogen bond.

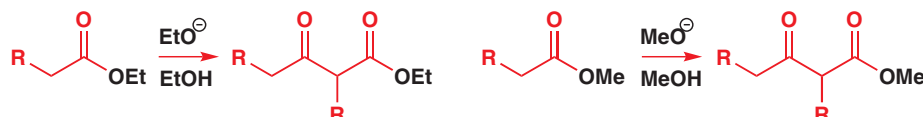


Summary of the preparation of keto-esters by the Claisen reaction

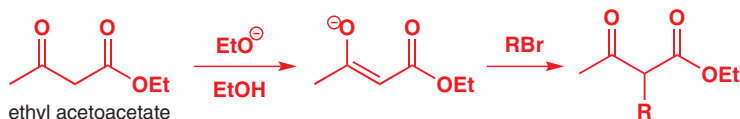
It is worth pausing at this moment to summarize which keto-esters can be made easily by the two methods we have discussed, namely

- Claisen ester condensation
- acylation of ketones with carbonates.

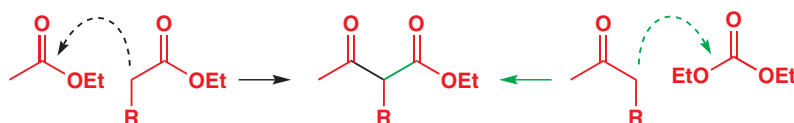
Ethyl acetoacetate (ethyl 3-oxobutyrates) can of course be made by the self-condensation of ethyl acetate. This ester is cheap to buy but homologues, available by the self-condensation of other esters, are usually made in the laboratory. Which esterifying group is used (OEt, OMe, etc.) is not important as long as the same alkoxide is used as the base.



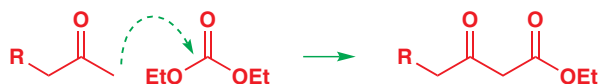
Compounds with only one of the R substituents in this structure are also easy to make. If the R substituent is at C2, it is best introduced by alkylation of the unsubstituted ester (see Chapter 25, p. 595).



Attempts to make this compound by the Claisen ester condensation would require one of the approaches in the diagram below. The dashed curly arrows suggest the general direction of the condensation required and the coloured bonds are those that would be formed *if* the reaction worked.

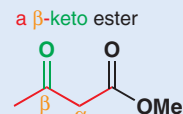


But neither reaction will work! The black route requires a controlled condensation between two different enolizable esters—a recipe for a mixture of products. The simple alkylation route above removes the need for control. The green route requires a condensation between an unsymmetrical ketone and diethyl carbonate. This condensation will work all right, but not to give this product. As you saw on p. 645, Claisen condensations prefer to give the less substituted dicarbonyl compound, and condensation would occur at the methyl group of the ketone on the right to give the other unsymmetrical keto-ester. So this isomer can be made easily too.



β -Keto esters

We pointed out on p. 643 that the Claisen reaction generates 1,3-dicarbonyl compounds. The examples here are a subset of such compounds: they are all β -keto esters:



• Making β keto-esters: a checklist

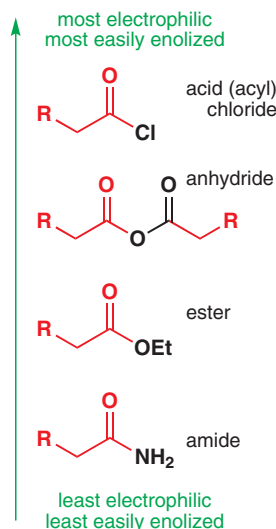
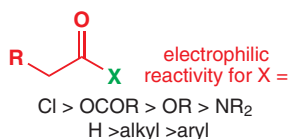
A combination of self-condensation, condensation with diethyl carbonate, and alkylation of keto-esters prepared by one of these means will allow us to make most β keto-esters that we are likely to want. Look out for all the usual problems of enolate chemistry and if any of these is a problem, try an alkylation route.

- Will the right carbonyl compound enolize?
- If it is a ketone, will it enolize in the right way?
- Will the enolate react with the right acylation partner?

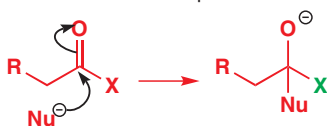
Controlling acylation with specific enol equivalents

In the first part of this chapter we saw how specific enol equivalents could be used to control aldol reactions. We now need to look at the same type of control in the acylation of enolates and extend our discussion to specific enolates of carboxylic acid derivatives.

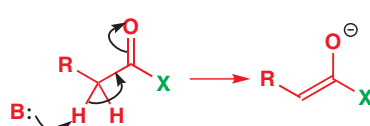
We established in Chapter 10 a hierarchy for the electrophilic reactivity of acid derivatives that should by now be very familiar to you—acyl chlorides at the top to amides at the bottom. But what about the reactivity of these same derivatives towards enolization at the position, that is, the CH_2 group between R and the carbonyl group in the various structures? You might by now be able to work this out. The principle is based on the mechanisms for the two processes.



mechanism of nucleophilic attack

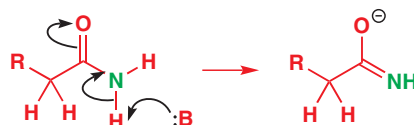
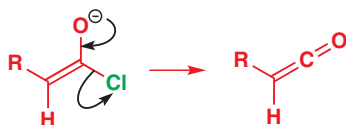


mechanism of enolate formation



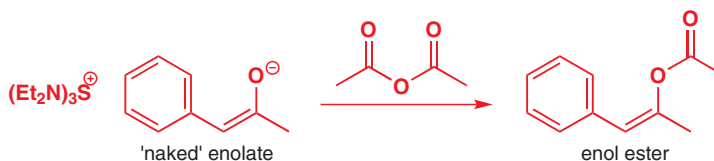
See how similar these two mechanisms are. In particular, they are the same at the carbonyl group itself. Electrons move into the $\text{C}=\text{O}$ π^* orbital: the $\text{C}=\text{O}$ bond becomes a $\text{C}-\text{O}$ single bond as a negative charge develops on the oxygen atom. It should come as no surprise that *the order of reactivity for enolization is the same as the order of reactivity towards nucleophilic attack*. Aldehydes are more electrophilic and more easily enolized than ketones and ketones are more electrophilic and more easily enolized than esters, although exact comparisons between aldehydes and ketones on the one hand and acid derivatives on the other are unwise.

In Chapter 20 we established that enolates can be formed from acid chlorides, but that they decompose to ketenes. Enolates can be formed from amides with difficulty, but with primary or secondary amides one of the NH protons is likely to be removed instead. For the remainder of this section we shall look at how to make specific enol equivalents of acids, esters, aldehydes, and ketones.

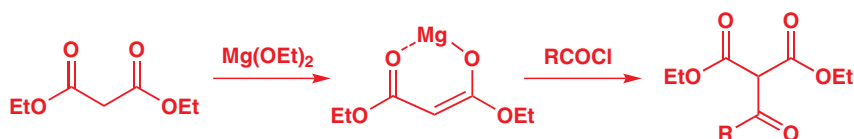


Directed C-acylation of esters

The danger we have to face is that acylation is inclined to occur on oxygen rather than on carbon. In the extreme case, naked enolates (those with completely non-coordinating cations) acylate cleanly on oxygen with anhydrides or acid chlorides.

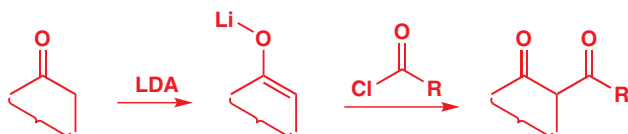


Fortunately, the reagents we have just discussed for aldol reactions (lithium and zinc enolates) are also acylated at carbon rather than on oxygen. Even with acid chlorides magnesium enolates, particularly those of 1,3-dicarbonyl compounds, give reliable C-acylation. The magnesium atom bonds strongly to both oxygens, lessening their effective negative charge.

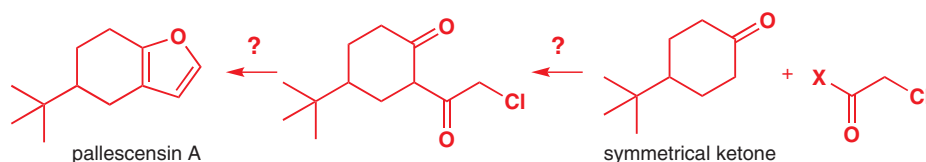


Hydrolysis and decarboxylation in the usual way lead to keto-esters or keto-acids. Of the more common metals used to form enolates, lithium is the most likely to give good C-acylation as, like magnesium, it forms a strong O–metal bond. It is possible to acylate simple lithium enolates with enolizable acid chlorides.

➔ See p. 597 for a discussion of decarboxylation.



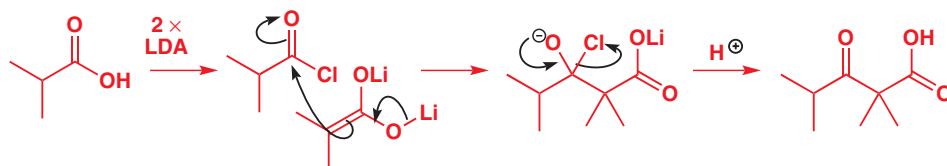
We shall describe two examples of this reaction being used as part of the synthesis of natural products. The first is pallescensin A, a metabolite of a sponge. It is quite a simple compound and some chemists in Milan conceived that it might be made from the chloro-diketone shown below, which might in turn be made by acylation of the enolate of a symmetrical ketone.



The route chosen was to react the lithium enolate of 4-*t*-butyl cyclohexanone with the correct acid chloride. This reaction worked well, as did the rest of the synthesis of pallescensin A, which was first made by this route. The key step, the acylation of the lithium enolate, is interesting because alkylation could have occurred instead. The acid chloride is more electrophilic than the alkyl chloride in this reaction, although alkylation does occur in the next step. Notice how the lithium atom holds the molecules together during the reaction.



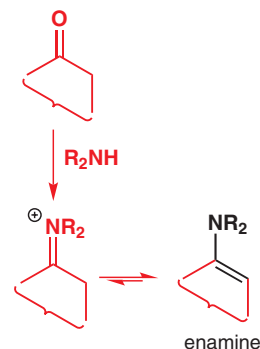
Even the dilithio derivatives of carboxylic acids, made by treating a carboxylic acid with two molecules of LDA, can give good reactions with acid chlorides. In these reactions it is not necessary to have a proton remaining between the two carbonyl groups of the product as the reaction is between a strong nucleophile and a strong electrophile and is under kinetic control.



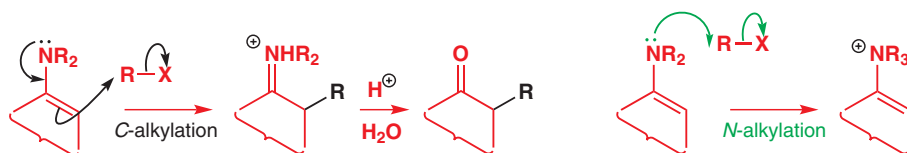
It is rather more common to use enamines or silyl enol ethers in acylations with acid chlorides. These are more general methods— enamines work well for aldehydes and ketones while silyl enol ethers work for all classes of carbonyl compounds. It is possible to combine two enolizable molecules quite specifically by these methods, and we shall consider them next.

The acylation of ketones via enamines and aza-enolates

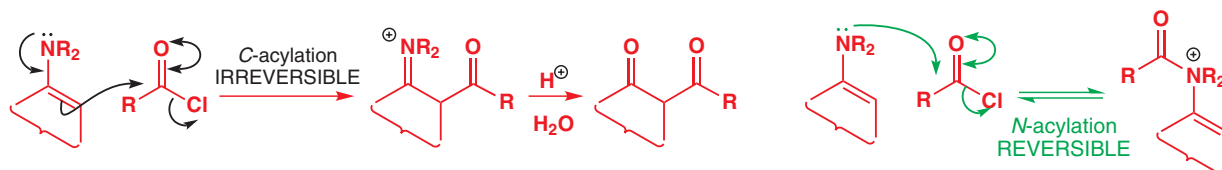
Enamines are made from secondary amines and aldehydes or ketones via the iminium salt: you met them in Chapter 11 and have seen them in action in Chapters 20 and 25. In Chapter 25 we saw that reliable C-alkylation of enamines occurs with reactive allyl halides and



α -halocarbonyl compounds, but that unwanted *N*-alkylation often competes with simple alkyl halides. We also noted earlier in this chapter that they are rarely used for aldol reactions as they are not reactive enough.

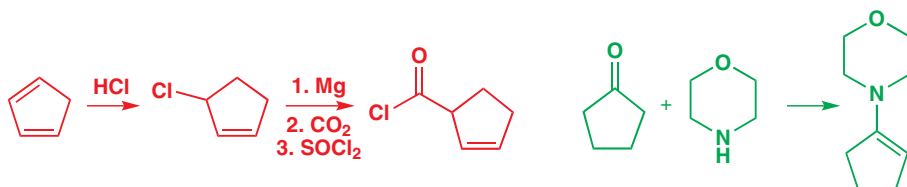


Acylation with the much more reactive acid chlorides could follow the same two pathways, but with one big difference. The products of *N*-acylation are unstable salts and *N*-acylation is reversible. Acylation on carbon, on the other hand, is irreversible. For this reason enamines end up acylated reliably on carbon.

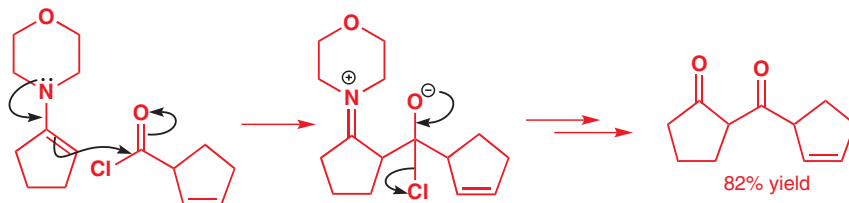


The Swiss chemist Oppolzer used just such a reaction in a synthesis of the natural product longifolene. He first prepared an acid chloride from cyclopentadiene, and the enamine from cyclopentanone and the secondary amine morpholine.

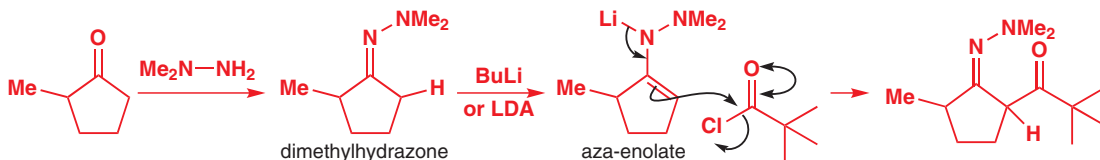
■ Morpholine is frequently used in the preparation of enamines. See p. 592.



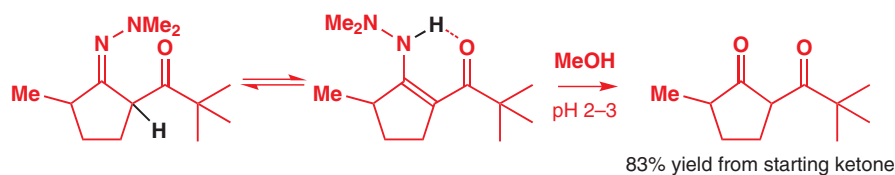
Combining the enamine with the acid chloride led to a clean acylation at carbon in 82% yield and eventually to a successful synthesis of longifolene.



Aza-enolates also react cleanly at carbon with acid chlorides. Good examples come from dimethylhydrazones of ketones. When the ketone is unsymmetrical, the aza-enolate forms on the less substituted side, even when the distinction is between primary and secondary carbons. The best of our previous regioselective acylations have distinguished only methyl from more highly substituted carbon atoms.



You will not be surprised to find that the immediate product tautomerizes to an acyl-enamine, further stabilized by an internal hydrogen bond. Mild acidic work-up releases the diketone product. The overall procedure may sound complicated— Me_2NNH_2 then base then acyl chloride then acidic methanol—but it is performed in a single flask and the products, the 1,3-diketones, are formed in excellent yield—in this case 83% overall.

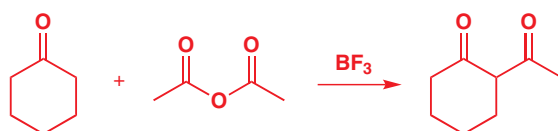


Hydrazone, as we explained on p. 232 of Chapter 11, are much less electrophilic than ketones. Even BuLi can be used as a base: it does not attack the $\text{C}=\text{N}$ bond.

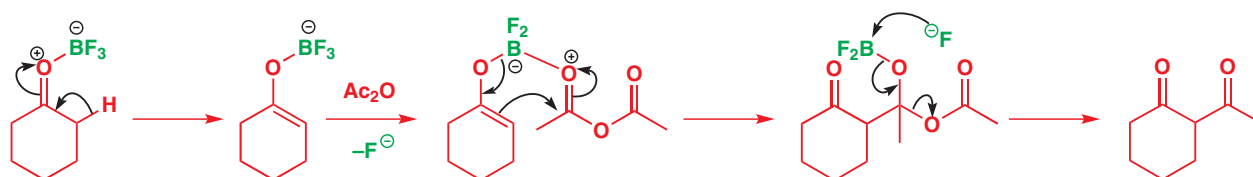
Interactive mechanism for hydrazone enolate alkylation

Acylation of ketones under acidic conditions

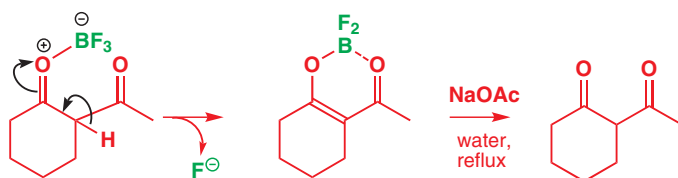
Acylation of ketone enols with anhydrides is catalysed by Lewis acids such as BF_3 . This process will remind you of Friedel–Crafts acylation (p. 477) but a better analogy is perhaps the aldol reaction, where metals such as lithium hold the reagents together so that reaction can occur around a six-membered ring.



The mechanism obviously involves attack by the enol (or 'boron enolate') of the ketone on the anhydride, catalysed by the Lewis acid. Probably the boron atom holds the reagents together, much as the lithium atom does in aldol reactions of lithium enolates (p. 625).

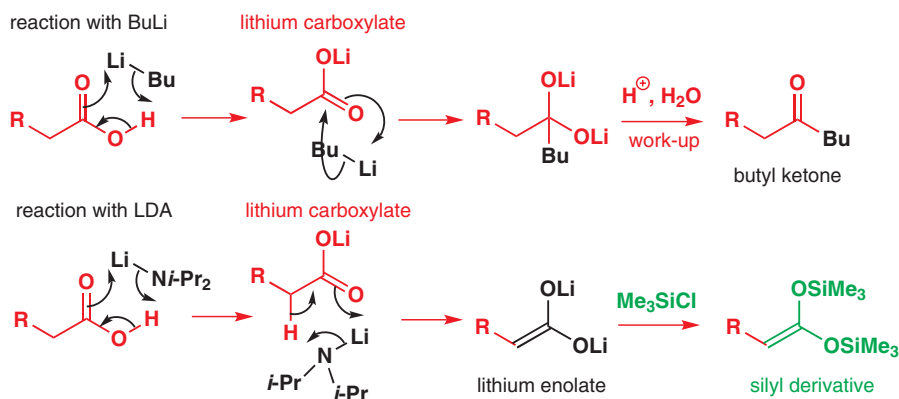


Under the conditions of the reaction, the product forms a stable boron enolate, which needs to be decomposed to the diketone with refluxing aqueous sodium acetate.



Acylation of free carboxylic acids

You might think that the presence of the acidic proton in a carboxylic acid would present an insuperable barrier to the formation and use of any enol derivatives. In fact, this is not a problem with either the lithium enolates or the silyl enol ethers. Addition of BuLi or LDA to a carboxylic acid immediately results in the removal of the acidic proton and the formation of the lithium salt of the carboxylic acid. If BuLi is used, the next step is addition of BuLi to the carbonyl group and the eventual formation of a ketone (see Chapter 10, p. 218). But, if LDA is used, it is possible to form the lithium enolate of the lithium derivative of the carboxylic acid.



■ Silyl enol ethers of acids or esters are called silyl ketene acetals. See p. 609 for more on this.

The enolate derivative is rather strange as it has two OLi groups on the same double bond, but it can be cleanly converted to the corresponding silyl enol ether. Both lithium enolates and silyl enol ethers from acids can be used in aldol reactions.

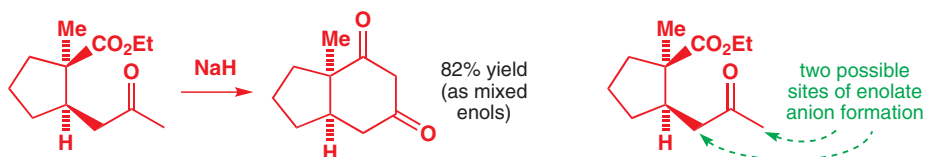
● Useful enolates for the aldol reaction and for acylation at carbon

Enolate type	Aldehyde	Ketone	Ester	Acid
lithium enolate	×	✓	✓	✓
silyl enol ether	✓	✓	✓	✓
enamine	✓	✓	×	×
aza-enolate	✓	✓	×	×
zinc enolate	×	×	✓	×

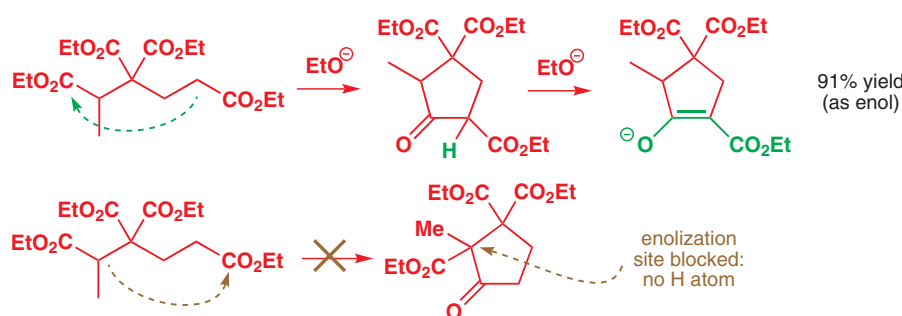
This concludes our general survey of specific enolates in acylation at carbon. We are left with some reactions that are particularly easy to do.

Intramolecular crossed Claisen ester condensations

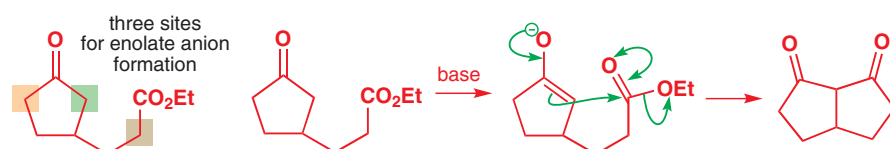
In the same way as with intramolecular aldol condensations, we do not have to worry so much about controlling where enolization occurs providing that one product is more stable than the others—for example, it might have a five- or a six-membered ring (rather than a four- or eight-membered one)—and we carry out the reaction under equilibrating conditions. A couple of examples should show what we mean. Although there are two sites for enolate anion formation, one would give a four-membered ring and can be ignored. Only enolization of the methyl group leads to a stable six-membered ring.



In this next example the two possible sites for enolate anion formation would both lead to stable five-membered rings. The product forms a stable enolate under the reaction conditions but the alternative cannot form a stable enolate as there is no hydrogen atom between the two carbonyl groups.



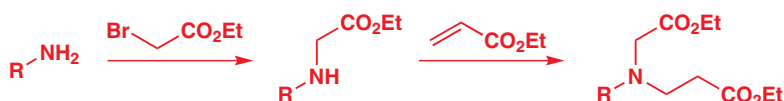
In the next example, there are three possible sites for enolate anion formation, but only one product is formed and in good yield too. If we consider all three possible enolate anions, the choice is more easily made. First, the reaction that *does* happen. An enolate anion is formed from the ketone at the green site and acylation at carbon follows. The product is a fused rather than a bridged bicyclic structure and can easily form a stable enolate anion.



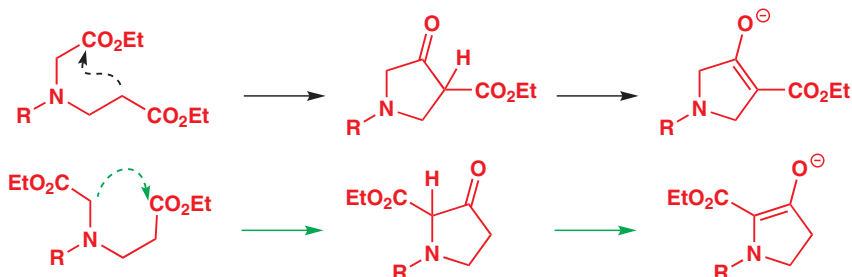
We could form the enolate anion on the other side of the ketone at the orange site and attack the ester in the same way. The product would be a bridged bicyclic diketone, and is not formed (see above). The third possible enolate site (brown) could give an aldol reaction but the product would again be a bridged bicyclic compound and is not formed.

Symmetry in intramolecular crossed Claisen condensations

If cyclization is to be followed by decarboxylation, a cunning plan can be set in motion. Addition of an amine by an S_N2 reaction to an α halo-ester followed by conjugate addition to an unsaturated ester gives a substrate for Claisen ester cyclization.



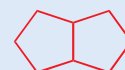
This diester is unsymmetrical so cyclization is likely to lead to two different keto-esters. Either can form a stable enolate so both are indeed formed. This sounds like very bad news since it gives a mixture of products.



The cunning plan is that the relative positions of the ketone and the nitrogen atom in the five-membered ring are the same in both products. All that differs is the position of the

Bicyclic compounds

In Chapter 32 we will discuss the differences between fused compounds (one bond in common), *spiro* compounds (one atom in common), and bridged compounds (rings joined at two non-adjacent atoms). Each of these three examples has two five-membered rings.



fused bicyclic

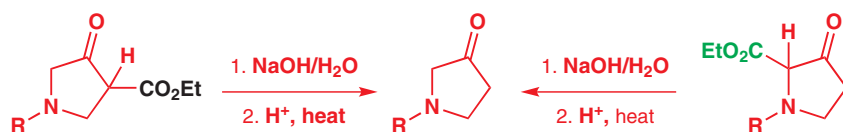


spirocyclic

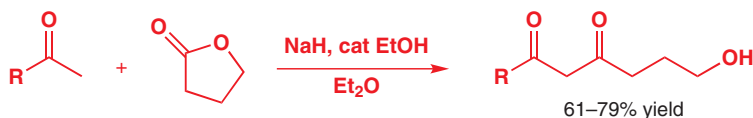


bridged bicyclic

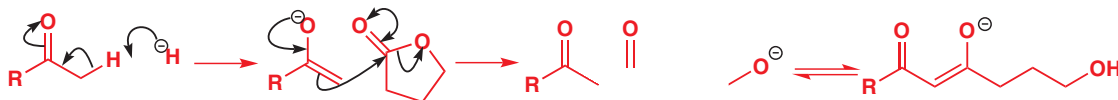
CO₂Et group. When the two different products are hydrolysed and decarboxylated they give the same amino-ketone!



Just occasionally it is possible to carry out cross-condensations between two different enolizable molecules under equilibrating conditions. A notable example is the base-catalysed reaction between methyl ketones and lactones. With sodium hydride—a strong base that can convert either starting material entirely into its enolate anion—good yields of products from the attack of the enolate of the ketone on the electrophilic lactone can be obtained.



Kinetic enolate formation must occur at the methyl group of the ketone followed by acylation with the lactone. Lactones are rather more electrophilic than non-cyclic esters, but the control in this sequence is still remarkable. Notice how a stable enolate is formed by proton transfer within the first-formed product.



Carbonyl chemistry—where next?

This chapter concludes a survey of the reactions of carbonyl compounds which started way back in Chapter 6 with an introduction to addition reactions at the C=O group, and moved on through the following stages:

- Chapter 9: C–C bonds by adding organometallics to C=O
- Chapter 10: Substitution at C=O (carboxylic acid derivatives)
- Chapter 11: Substitution at C=O with loss of the carbonyl O (acetals, imines, etc.)
- Chapter 20: Enols and enolates
- Chapter 25: Alkylating enolates
- Chapter 26: Adding enols and enolates to C=O groups: the aldol and Claisen reactions

Carbonyl groups are the ‘hooks’ that allow chemists to put molecules together, and in the chapter after next (Chapter 28) we will discuss how we think about the science of synthesis using carbonyl reactivity. We will revisit many of the reactions you have seen not just in this next chapter, but beyond—in particular in the synthesis of heterocycles (Chapter 30) and in diastereoselective and enantioselective reactions (Chapters 33 and 41).

Further reading

S. Warren, *Chemistry of the Carbonyl Group*, Wiley, Chichester, 1974: section 5 is ‘Building Organic Molecules from Carbonyl Compounds.’ More advanced treatment: P. Wyatt and S. Warren, *Organic Synthesis: Strategy and Control*, Wiley, Chichester, 2007, chapters 3–6. The ultimate source for all types of aldol reaction is A. T. Nielsen and W. J. Houlihan, *Organic Reactions*, 1968, 16, whole volume. And for the

Claisen style condensations, J. P. Schaefer and J. J. Bloomfield, *The Dieckmann Condensation*: G. Jones, *The Knoevenagel Condensation: Organic Reactions*, 1967, 15, whole volume.

F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry B, Reactions and Synthesis*, 5th edn, Springer 2007, chapter 2.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>